Climate Shifters in a Structural Bioeconomic Model: Distributional Effects on Chilean Small Pelagic Fisheries

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Abstract

In Chile's multi-species small pelagic fishery—anchoveta, sardina común, and jack mackerel—climate change reshapes long-run stock productivity, with heterogeneous incidence across the artisanal and industrial fleets that target these stocks under the country's *Límite Máximo de Captura por Armador* (LMCA) quota allocation regime. We identify a set of structural climate parameters $(\rho_i^{SST}, \rho_i^{CHL})$ — semi-elasticities of intrinsic stock productivity with respect to a one-degree Celsius sea-surface temperature anomaly and a one-log-point chlorophyll-a anomaly, respectively — within a Bayesian state-space specification of the biological law of motion calibrated on official IFOP and SPRFMO assessments for 2000–2024. Priors are derived from a transparent reduced-form stress-test protocol over the same period, and a complementary negative binomial model captures annual fishing trips as a function of prices, vessel-level harvest allocations, weather, and regulatory closures, estimated separately for the artisanal and industrial fleets. Applying the identified shifters to a six-model CMIP6 ensemble (IPSL-CM6A-LR, GFDL-ESM4, CESM2, CNRM-ESM2-1, UKESM1-0-LL, MPI-ESM1-2-HR; SSP2-4.5 and SSP5-8.5; mid- and end-of-century), we report the comparative-statics change in intrinsic productivity as a full posterior with cross-model and within-model uncertainty separated explicitly: anchoveta's cross-model median declines by 45% under SSP2-4.5 mid-century and by 90% under SSP5-8.5 end-of-century; sardina común declines by 80% and 99.9%, respectively, with the posterior probability of negative impact equal to one in every scenario–window pair for both stocks. The jack mackerel shifters are not identified—the posterior-to-prior standard-deviation ratio is essentially unity, against 0.4–0.8 for the coastal-upwelling stocks—consistent with the species' transboundary SPRFMO management. Because the artisanal fleet's portfolio is concentrated in the two coastal-upwelling stocks while the industrial fleet is diversified by its allocation to jack mackerel, Chile's

Límite Máximo de Captura por Armador (LMCA) regime — with limited cross-sector quota transferability — translates the climate signal into a disproportionate long-run loss of harvest capacity for the artisanal segment. This carries direct implications for the design of the LMCA under non-stationary climate.

Keywords: climate change; small pelagic fisheries; Bayesian state-space; quota allocation; multi-species; distributional effects.

1 Introduction

Climate change is reshaping the productivity of marine fisheries with heterogeneous incidence across fleet segments, raising first-order questions for the design of quota-allocation regimes that were originally calibrated under stationary climate assumptions. The distribution and abundance of marine resources are changing in response to environmental conditions such as global ocean warming (Poloczanska et al., 2013), and these biophysical shifts translate into economic consequences for fishers that depend on the differential exposure of each fleet's species portfolio to the climate signal and on the institutional architecture that governs the cross-sector reallocation of fishing rights. Climate change is projected to shift species distributions, reducing availability in some areas while increasing it in others, with direct consequences for fishing communities (Sumaila et al., 2011). Understanding how these changes will propagate to fishing activity is a prerequisite for designing effective adaptation policies. Adaptation responses by fishers are heterogeneous and depend on institutional, economic, and ecological context (Jardine et al., 2020; Zhang & Smith, 2011), which motivates empirical analyses at the local level.

The Chilean small pelagic fishery (SPF) provides a particularly informative setting. It is the largest fishery in the country, accounting for almost 94% of total Chilean landings in 2019 (SUBPESCA, 2020), and is dominated by three species—anchoveta (*Engraulis ringens*), sardine (*Strangomera bentincki*), and jack mackerel (*Trachurus murphyi*)—that share habitat and are coupled through both trophic and market linkages. Cheung et al. (2010) project Chile among the top fishing nations in terms of declines in maximum catch potential by mid-century, with an estimated reduction of 6–13% under the SRES A1B scenario. Yet the Southeast Pacific remains one of the least studied regions regarding the economic impacts of climate change on fisheries (Sumaila et al., 2011), and, to our knowledge, no existing study for Chile combines multi-species stock dynamics with vessel-level effort responses under projected climate scenarios.[1]

A multi-species framework is essential in this setting because the three target species are ecologically and economically interdependent. Jack mackerel preys on anchoveta and sardine, so reductions in anchoveta or sardine availability may reflect not only direct environmental drivers

[1]For Chile, the only empirical study of fisher behavior using discrete choice modeling that we are aware of is Peña-Torres et al. (2017), which analyses how the El Niño–Southern Oscillation (ENSO) affects location choices in the jack mackerel fishery.

but also changes in predation pressure (Alheit & Niquen, 2004; Arancibia et al., 2019). Together with a fleet structure in which the industrial and artisanal segments target different species, this generates cross-fleet spillovers through biology, quota allocation rules, and shared port infrastructure. Current management—which sets the TAC on a single-species basis—does not internalize these interdependencies, strengthening the case for a multi-species approach.

We contribute to the literature in three ways. First, we identify—rather than merely estimate—a set of structural climate shifters that modulate the intrinsic growth rate of each stock, within a Bayesian state-space specification of stock dynamics calibrated on official IFOP and SPRFMO assessments. The identified parameters are interpretable as semi-elasticities of stock productivity with respect to sea-surface temperature and log-chlorophyll anomalies, and are invariant to the joint historical distribution of biomass, harvest, and environmental conditions. Second, we couple this structural model with a negative binomial model for annual fishing trips estimated separately for the industrial and artisanal fleets, and we evaluate the identified shifters at CMIP6 projections from a six-model ensemble (IPSL-CM6A-LR, GFDL-ESM4, CESM2, CNRM-ESM2-1, UKESM1-0-LL, MPI-ESM1-2-HR) under SSP2-4.5 and SSP5-8.5 to obtain posterior distributions of the long-run comparative-statics change in each stock's intrinsic productivity at mid-century (2041–2060) and end-of-century (2081–2100), with cross-model and within-model uncertainty separated explicitly. Third, we link the identified structural response to the negative binomial trip equations estimated separately for each fleet and show that the long-run distributional incidence of climate change across the artisanal and industrial segments is governed by the interaction between (i) the differential exposure of each fleet's species portfolio to the identified shifters and (ii) the limited cross-sector transferability built into the LMCA quota regime. The artisanal fleet, concentrated in the two coastal-upwelling stocks whose climate semi-elasticities are identified and large, bears a sharper long-run decline in harvest capacity than the industrial fleet, whose portfolio is diversified by jack mackerel — a mechanism consistent with the broader fishery portfolio literature on income diversification and risk (Cline et al., 2017; Kasperski & Holland, 2013) and with evidence that institutional access constraints can limit the realized benefits of diversification (Oken et al., 2021).

2 The small pelagic fishery in Chile

The Centro-Sur Chilean coast lies within the Humboldt Current System, an eastern boundary upwelling system in which prevailing southerly winds drive cold, nutrient-rich deep water to the surface near the coast. This wind-driven upwelling sustains one of the most productive marine ecosystems in the world, and the three target species of the Chilean small pelagic fishery — anchoveta, sardina común, and jack mackerel — occupy this productive coastal corridor. Climate change affects this system primarily through warming of the surface ocean (which the species perceive directly through their thermal envelope) and through changes in the wind-driven upwelling regime that modulate the supply of nutrients and therefore primary productivity at the base of the

food web. The first channel is captured in the present paper through the sea-surface temperature shifter ρ^{SST}, the second through the chlorophyll-a shifter ρ^{CHL}.

The Chilean small pelagic fishery is structured into two latitudinal zones with distinct species composition and fleet activity. In the northern zone competition is largely between anchoveta and jack mackerel, while in the Central-South region (Regions V–X and XIV) all three species play a major role. The Central-South is therefore the most relevant setting for the study of species interactions and potential substitution within a multispecies management framework, and it is the focus of this paper.

The jack mackerel fishery was initially concentrated in northern Chile but shifted to Central-South Chile in the mid-1980s, where the main fishing grounds have traditionally operated within 50 nautical miles of the coast (Peña-Torres et al., 2017). Landings of the three species have historically been destined primarily for the reduction industry: between 1987 and 2004, approximately 85% of jack mackerel landings were processed into fishmeal and fish oil (Peña-Torres et al., 2017). Today, the main ports servicing the SPF include San Antonio, Tomé, Talcahuano, San Vicente, Coronel, Lota, and Corral. Sardine and anchoveta are harvested as a mixed fishery in the Central-South region: the two species share the same habitat and are caught with the same gear, making differentiation at the point of capture effectively impossible (Dresdner et al., 2013).

2.1 Stock status

Central-South anchoveta was classified as collapsed through 2018, shifted to overexploited status in 2019, and has been fished within maximum sustainable yield (MSY) limits since 2020. Sardine has generally remained within MSY limits, except in 2021 and 2023 when it was classified as overexploited. Jack mackerel was overexploited until 2018 and has since been harvested within MSY limits.

2.2 Management regime

The Chilean SPF is managed through an annual Total Allowable Catch (TAC, *Cuota Global*) divided between the industrial and artisanal sectors, with small shares reserved for research and contingency. The TAC is subdivided by region and season, and unused quotas can be reassigned within the fishing year. Anchoveta and sardine are regulated as a mixed-species fishery—each species has its own quota, but substitution between them is permitted.

Since 2013, the industrial sector has operated under an individual transferable quota (ITQ) system known as Transferable Fishing Licenses (*Licencias Transables de Pesca*, LTP). Class A licenses were allocated on the basis of historical catches, while Class B licenses (up to 15% of the industrial fraction) are allocated through sealed-bid first-price auctions that began in 2015. Peña-Torres et al. (2022) document that these auctions have exhibited low participation, limited ability

to reflect economies of scale, and signs of coordinated bidding. The artisanal sector operates under a regulated freedom-to-fish regime, with regional TACs allocated through the *Régimen Artesanal de Extracción* (RAE) by area, vessel size, landing site, or fisher organization.

Biological closures (*vedas*) are a central regulatory instrument for sardine and anchoveta. In the Central-South region, reproductive closures run from July–October in regions IX–XIV and from August–October in regions V–VIII, while recruitment closures run from January to February across all regions. In total these yield 151 closed days per year in regions V–VIII and 182 closed days in regions IX–XIV. Jack mackerel has no seasonal biological closures.

3 Data and methodology

Our empirical approach has three stages. First, we identify a multi-species stock dynamics model using a Bayesian state-space specification on the latent biomass of each stock, with a Schaefer surplus-production function whose intrinsic growth rate is modulated by an interpretable climate shifter. Second, we estimate a negative binomial model for annual fishing trips as a function of prices, harvest allocations, weather, and regulatory closures, separately for the artisanal and industrial fleets. Third, we evaluate the identified shifters under CMIP6 projections from a six-model ensemble to assess how climate change will reshape the long-run productivity of each stock and, through the trip equation, the fleet-level allocation of fishing effort, with cross-model and within-model uncertainty separated explicitly.

3.1 Historical data

We use data requested from the Chilean Fisheries Development Institute (Instituto de Fomento Pesquero, IFOP) covering the 2013–2024 period. The dataset includes trip-level microdata, which contain detailed records on vessel identifiers, departure and arrival times, vessel capacity, fleet and gear type, ports of departure and landing, fishery codes, haul timing and location, species composition, retained catch, and trip activity. In addition, we requested annual information on stock abundance and vessel landings by port, county, region, country, and species. Finally, we use ex-vessel prices, reported monthly or annually by port, county, region, country, and species. These prices reflect those paid by processing plants to fishers at the point of first sale and are obtained through IFOP's landing surveys, which do not necessarily cover all market transactions.

For the environmental covariates, we use data from the E.U. Copernicus Marine Service Information, accessed through the Copernicus Marine Toolbox API. Salinity, sea surface temperature, and current speed and direction were obtained from the Global Ocean Physics Reanalysis (GLORYS12V1), which provides data at a 1/12° horizontal resolution with 50 vertical levels (E.U. Copernicus Marine Service Information, 2025c). Wind speed and direction at the surface were obtained from the Global Ocean Hourly Reprocessed Sea Surface Wind and Stress from Scatterometer and

Model dataset, available at 0.125° horizontal spatial resolution and hourly frequency (E.U. Copernicus Marine Service Information, 2025b). Chlorophyll-a concentrations were obtained from the Global Ocean Colour dataset, which provides data at ~4 km horizontal resolution (E.U. Copernicus Marine Service Information, 2025a). All environmental data were retrieved daily (hourly in the case of winds) for the 2013–2024 period, covering the Chilean Exclusive Economic Zone (EEZ) between 32°S and 41°S (Figure 1)

Observation structure for jack mackerel Centro-Sur biomass Over the 2000–2024 estimation window (25 years) the IFOP hydroacoustic assessment of jack mackerel in Centro-Sur Chile provides 16 years with a point estimate of spawning biomass and two further years (2012, 2015) in which biomass fell below the assessment's lower detection limit and enters the likelihood as a left-censored observation at that limit. The remaining seven years were either not surveyed or lacked sufficient information for a stock-assessment run; in the state-space specification of Section 3.3.1 these years carry no observation and their latent biomass $B_{\text{jurel},t}$ is identified dynamically, through the Schaefer transition equation together with the stock-specific priors on $(r^0, K, \sigma_{\text{proc}}, \sigma_{\text{obs}})$ adopted from the IFOP and SPRFMO assessments and on (ρ^{SST}, ρ^{CHL}) adopted from the reduced-form stress tests of Appendix A.

Two external series—jack mackerel biomass from the Northern acoustic zone (Arica–Iquique–Antofagasta, available since 2010) and the SPRFMO transzonal spawning biomass index for the Southeast Pacific—were considered as auxiliary information. The Northern series correlates with the observed CS series at $r = 0.82$ over $N = 7$ overlapping years, and the SPRFMO index correlates with CS at $r = 0.11$ over $N = 17$ overlapping years. We read these numbers as suggestive, not conclusive: a seven-point Pearson correlation is at best descriptive, but the sign and magnitude contrast is consistent with a Centro-Sur stock unit that shares short-run dynamics with the Northern Chilean assessment and is only weakly coupled to the SPRFMO-scale aggregate. Neither external series enters the likelihood of the structural model, as a prior or as an observation, to avoid introducing information about the coastal climate forcing through the indirect route of Northern or transzonal biomass.

3.2 Future climate data

Climate projections for SST, chlorophyll-a, and near-surface wind speed are obtained from the CMIP6 archive, using a six-model ensemble (IPSL-CM6A-LR, GFDL-ESM4, CESM2, CNRM-ESM2-1, UKESM1-0-LL, MPI-ESM1-2-HR) selected to span the CMIP6 equilibrium climate sensitivity distribution. We use monthly outputs for two future scenarios—SSP2-4.5 (moderate) and SSP5-8.5 (high emissions). The projection horizons, baseline period, and delta-method procedure used to translate CMIP6 outputs into our empirical models are presented in Section 3.4; the per-model decomposition of projection uncertainty appears in Appendix F for stock-level intrinsic productivity and in Appendix G for fleet-level trips.

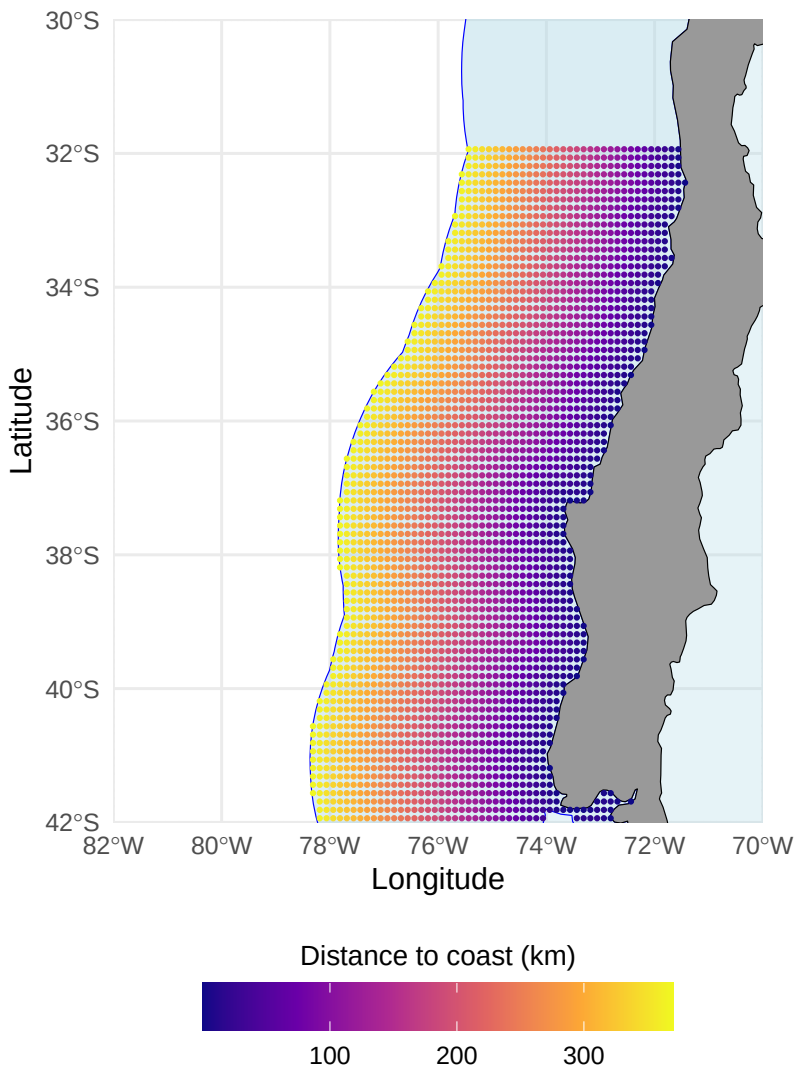

Figure 1: Geographical extent of the study area used for environmental covariates limited to the Chilean Exclusive Economic Zone

3.3 Econometric models

3.3.1 Stock dynamics

We model the interannual dynamics of each stock $i \in \{\text{anchoveta}, \text{sardina común}, \text{jack mackerel}\}$ as a Bayesian state-space process on the latent biomass $B_{i,t}$, driven by a Schaefer surplus-production function whose intrinsic growth rate is modulated by the physical climate. Schaefer is adopted as the $\theta = 1$ specialization of the Pella–Tomlinson family on identifiability grounds: the shape parameter θ is notoriously weakly identified with $N = 25$ annual observations per stock, and imposing $\theta = 1$ is a standard identification device in the small-pelagic literature (Hilborn & Walters, 1992; Maunder & Punt, 2013). The law of motion is

$$B_{i,t+1} \;=\; B_{i,t} + r_{i,t}\, B_{i,t}\left(1 - \frac{B_{i,t}}{K_i}\right) \;-\; C_{i,t} \;+\; \varepsilon_{i,t}, \qquad \varepsilon_{i,t} \sim \mathcal{N}(0,\, \sigma^2_{\text{proc},i}), \tag{1}$$

where $C_{i,t}$ is the observed annual catch, K_i is the stock-specific carrying capacity, and $\sigma_{\text{proc},i}$ is the standard deviation of the stochastic recruitment innovation. Climate enters the dynamics through a log-linear shifter on the intrinsic growth rate,

$$r_{i,t} \;=\; r_i^0 \,\exp\!\Big(\, \rho_i^{SST}\,(SST_{t-1} - \overline{SST}) \;+\; \rho_i^{CHL}\,(\log CHL_{t-1} - \overline{\log CHL})\,\Big), \tag{2}$$

so that ρ_i^{SST} and ρ_i^{CHL} are the semi-elasticities of stock productivity with respect to a one-degree Celsius SST anomaly and a one-log-point chlorophyll-a anomaly, respectively. The vector $(\rho_i^{SST}, \rho_i^{CHL})_{i=1}^3$ is the structural object on which Section 4.1 focuses. Because these parameters enter the law of motion as coefficients on exogenous climate forcings, they are invariant to the joint historical distribution of (SST, B, C) and therefore transportable to climate regimes outside the 2000–2024 estimation sample. This invariance is the substantive justification for the structural specification, given that the comparative-statics exercise of Section 4.3 evaluates the climate shifter at projected SST anomalies that lie several historical standard deviations beyond the estimation envelope.

Latent biomass is linked to the official survey-based estimates of spawning stock biomass $B_{i,t}^{\text{obs}}$—published by IFOP for anchoveta and sardina común in Centro-Sur Chile and by SPRFMO for jack mackerel—through a log-normal measurement equation,

$$\log B_{i,t}^{\text{obs}} \;=\; \log B_{i,t} \;+\; u_{i,t}, \qquad u_{i,t} \sim \mathcal{N}(0,\, \sigma^2_{\text{obs},i}). \tag{3}$$

Environmental conditions are summarized annually by sea surface temperature (SST) and chlorophyll-a concentration (CHL), averaged over the Centro-Sur region within the Chilean EEZ and lagged one year to respect the pre-determined ordering of environmental forcing relative to year-t biomass. The choice of these two covariates follows a large biological literature that identifies thermal regime and primary productivity as the dominant local drivers of small-pelagic dynamics in the Humboldt Current System (Axbard, 2016; Cahuin et al., 2009; Yáñez et al., 2014; see Cheung et al., 2008; Jennings et al., 2008).

The state-space specification is estimated by Bayesian inference rather than maximum likelihood for two reasons that are standard in the small-sample fishery literature: with $N = 25$ annual observations per stock, the joint identification of the biological parameters (r_i^0, K_i) and the climate shifters $(\rho_i^{SST}, \rho_i^{CHL})$ requires the prior information embedded in the IFOP and SPRFMO single-species assessments to avoid degenerate likelihood maxima at the boundary of the parameter space (see Appendix A for an explicit demonstration of this boundary problem under maximum likelihood); and posterior credible intervals provide a natural framework for propagating the uncertainty

in the climate shifters through to the comparative-statics exercise of Section 4.3, which would otherwise require a non-trivial delta-method approximation around an unstable point estimate. Readers familiar with the maximum likelihood treatment of state-space models in the macro-econometrics literature (e.g., Kalman filtering with EM updates) will find the Bayesian treatment a direct generalisation, where the EM step is replaced by the joint posterior over latent states and parameters. The system is estimated by full-information Bayesian inference in Stan via Hamiltonian Monte Carlo. Priors on the biological parameters $(r_i^0, K_i, \sigma_{\text{proc},i}, \sigma_{\text{obs},i})$ are adopted from the single-species official assessments of IFOP and SPRFMO; priors on the climate shifters $(\rho_i^{SST}, \rho_i^{CHL})$ are weakly informative normal densities centred at the point estimates of an independent stress-test regression of biomass growth increments on SST and log-CHL anomalies, with unit standard deviation (see Appendix A for full specification of the prior elicitation protocol). We estimate three nested specifications of increasing flexibility: (i) species-independent dynamics without climate shifters (`ind''), (ii) with a cross-stock residual covariance $\Omega$ (omega''), and (iii) the full model augmented with the climate shifter of Eq. (2) (`full''). The posterior of thefull'' specification is our primary object of interpretation; the nested predictive comparison is relegated to Appendix B. All reported posteriors are based on four independent Hamiltonian Monte Carlo chains of 2,000 post-warmup iterations each, evaluated against standard convergence diagnostics: the Gelman–Rubin statistic $\hat{R}$ (which compares within-chain to between-chain variance and should be close to one if chains have mixed) is below 1.01 for all top-level parameters, and the effective sample size — both in the bulk of the posterior distribution and in its tails — exceeds 400 for all top-level parameters, indicating that posterior summaries are not contaminated by autocorrelation between adjacent draws. These thresholds are conventional in the applied Bayesian literature; see Appendix D for the convergence diagnostics by parameter family. The seven jack mackerel Centro-Sur years without a survey estimate enter the likelihood as unobserved latent biomass states, identified only through the Schaefer transition equation; two further years (2012 and 2015) enter through a log-normal censored-observation term at the stock-assessment detection limit. All associated uncertainty propagates directly into the posterior of $(\rho_{\text{jurel}}^{SST}, \rho_{\text{jurel}}^{CHL})$, so that no imputation step is embedded in the Stan likelihood.

Three features of the state-space formulation in Eq. (1)–(3) are material for this paper. First, it separates process noise σ_{proc} from observation noise σ_{obs}, which is essential given the documented coefficient-of-variation of acoustic and egg-production surveys in this fishery. Second, it enforces the pre-determined ordering between year-$(t-1)$ environmental forcing and year-t latent biomass, ruling out contemporaneous feedback into the climate covariates. Third, it yields structural climate semi-elasticities $(\rho_i^{SST}, \rho_i^{CHL})$ that are invariant to the joint historical distribution of (SST, B, C) and therefore transportable to climate regimes outside the 2000–2024 estimation window. Because the comparative-statics exercise of Section 4.3 evaluates the climate shifter at projected SST anomalies that lie several historical standard deviations beyond the estimation envelope, this transportability property is the substantive justification for the structural specification.

Figure 2 shows that anchoveta, sardina común, and jack mackerel exhibit distinct biomass tra-

jectories over the 2000–2024 window, with no evidence of strong interannual co-movement across species once differences in scale are accounted for. Harvest pressure is visibly associated with these fluctuations; jack mackerel, in particular, experienced an abrupt decline during the late 2000s consistent with the combined effects of intense fishing and unfavorable environmental conditions.

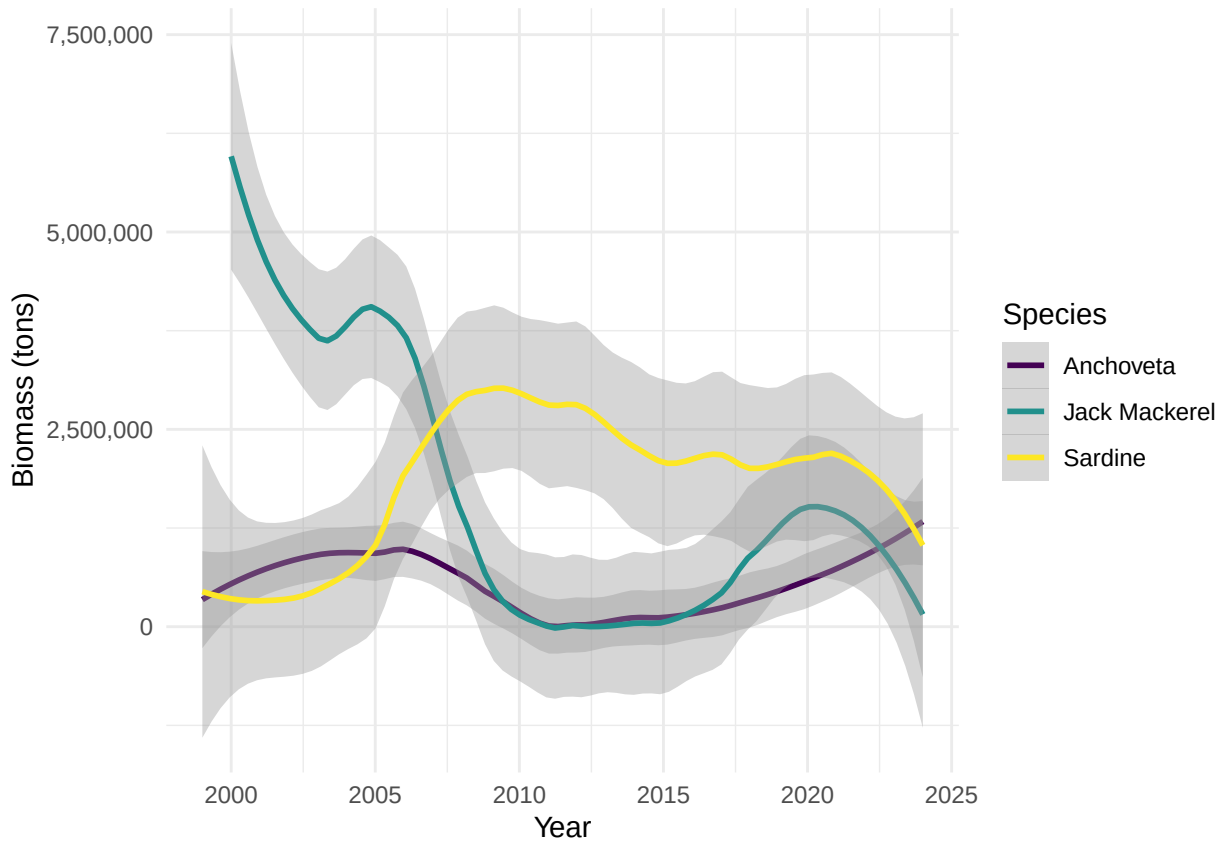

Figure 2: Estimated biomass of small pelagic species in Chile (2000–2024)

3.3.2 Total annual trips

We model the annual number of fishing trips taken by vessel v in year y as a count process following Kasperski (2015). Since the available logbook data only record purse-seine operations, the unit of observation is a vessel–year. Separate effort models are estimated for the industrial and artisanal fleets, allowing fishing activity to respond differently to economic conditions and regulatory constraints across sectors. This specification explicitly accommodates technological heterogeneity across fleet segments, reflecting differences in vessel capacity, operating scale, and production technology. A complementary approach, used in Quezada-Escalona et al. (2026) for the U.S. West Coast Coastal Pelagic Species fishery, models daily participation and target-species choice through a discrete choice framework with species distribution models as proxies for daily availability. The annual NB specification adopted here is the natural counterpart for a setting where vessel-year

is the operational unit at which TACs are allocated under the LMCA, and where daily logbook coverage is too sparse to support a daily DCM for the Centro-Sur Chilean SPF over the 2013–2024 window.

The baseline specification is a Poisson model:

$$T_{vy} \sim \text{Poisson}(\lambda_{vy}), \qquad \lambda_{vy} = \exp\big(U'_{vy}\beta\big)\,, \tag{4}$$

where T_{vy} denotes the total number of purse-seine trips recorded for vessel v in year y. Given substantial overdispersion in the data—the variance-to-mean ratio of T_{vy} is 22.4 for the artisanal fleet and 4.9 for the industrial fleet—we estimate a negative binomial (NB) model, which nests the Poisson as a special case. A likelihood ratio test strongly rejects the Poisson restriction in both fleets ($p < 0.001$).

The vector of explanatory variables U_{vy} includes output prices by species, vessel-level allocated harvest, fixed vessel characteristics, and operating conditions:

$$U_{vy} = [p_{sy},\, H_{vy}^{alloc},\, Z_v,\, O_{vy}]. \tag{5}$$

Output prices p_{sy} correspond to species-specific ex-vessel prices paid by processing plants to fishers, obtained from IFOP's manufacturing survey and deflated to constant 2018 pesos using the consumer price index from the Central Bank of Chile. Consistent with Dresdner et al. (2013), prices are measured as annual averages over peak fishing months to reflect the economically relevant conditions faced by vessels when planning annual effort.

Because the objective of the model is to characterize vessels' effort responses to changes in prices, quotas, and environmental conditions within the small pelagic fishery, we restrict attention to SPF trips recorded in logbooks. These trips account for approximately 95% of observed fishing revenue and therefore capture the primary economic margin through which vessels adjust annual fishing effort. Cross-validation against the SERNAPESCA vessel-level landings database (transparency request AH010T0006857; 2013–2024 Centro-Sur) confirms that the resulting panel is representative of the purse-seine artisanal segment. For the industrial fleet the panel covers 40 of 59 registered vessels (68% by count) and aggregate landings within −10% of the SERNAPESCA all-gear total, with portfolio-weighted catch composition matching SERNAPESCA aggregates within two percentage points across the three target species. For the artisanal fleet the panel covers 859 of 2{,}689 registered vessels (32% by count) but accounts for approximately 70% of total Centro-Sur artisanal landings of the three species, reflecting the concentration of catch in the larger purse-seine vessels for which logbook submission is mandatory. Smaller artisanal vessels using non-purse-seine gears (lampara, lines) operate outside the LMCA quota architecture and are not modeled in the trip equation; they appear in the SERNAPESCA aggregate landings used as the likelihood input for the biomass state-space model.

Quota shares do not enter Eq. (4) directly. Instead, shares translate annual TACs into vessel-level allocated harvests. Let ω_{vs}^{r} denote vessel v's historical share of landings of species s within its administrative region r (for artisanal vessels) or regulatory zone (for industrial vessels), computed over the full sample period. Given the effective TAC $\bar{Q}_{sy}^{r}$ for species s in year y in region r—obtained from SERNAPESCA quota monitoring records and reflecting all inter-fleet transfers and adjustments—vessel-level allocated harvest is:

$$H_{vy,s}^{alloc} = \omega_{vs}^{r}\,\bar{Q}_{sy}^{r}, \qquad H_{vy}^{alloc} = \sum_{s} H_{vy,s}^{alloc}. \tag{6}$$

Each vessel is assigned to its administrative region based on its modal departure port from the logbook records. Artisanal TACs are assigned at the regional level (regions V, VIII, IX, XIV, X), while industrial TACs follow the regulatory zone structure established by SUBPESCA (zones V–IX and XIV–X for jack mackerel; zone V–X for sardine and anchoveta). This regional construction introduces cross-sectional variation in H_{vy}^{alloc} that reflects the heterogeneous quota environments faced by vessels operating in different parts of the Centro-Sur fishery.

The vector Z_v contains time-invariant vessel characteristics, including hold capacity (log of cubic meters) and vessel type. Hold capacity determines the physical constraint on catch per trip and proxies for vessel scale.

The vector O_{vy} captures operating and regulatory constraints that vary across both vessels and years. Unlike the environmental variables in the stock dynamics model, which reflect biological productivity (SST, chlorophyll-a), the variables in the trip equation capture conditions that affect the *feasibility* of fishing operations. Two vessel–year variables are constructed using each vessel's center of gravity (COG)—the catch-weighted centroid of its haul locations over the sample period. First, the number of days with adverse weather conditions is computed as the count of days per year in which wind speed exceeds 8 m/s at the environmental grid point nearest to the vessel's COG.[2] Second, the number of days closed due to biological closures (*vedas*) is assigned based on the regulatory zone corresponding to the vessel's COG latitude. In the Centro-Sur region, sardine and anchoveta face seasonal closures for reproduction (August–October in regions V–VIII; July–October in regions IX–XIV) and recruitment (January–February in all regions), yielding 151 and 182 closed days per year, respectively. Jack mackerel has no biological closures.

Quota prices are not included explicitly in the trip equation. Instead, the economic scarcity of quota is captured implicitly through aggregate TAC levels and vessel-level allocated harvest.

Trips, harvest, and prices may be jointly determined within the year, implying potential endogeneity in reduced-form trip regressions. However, the purpose of Eq. (4) is not causal identification but rather to provide an empirically grounded behavioral relationship that maps changes in prices,

[2]The wind threshold of 8 m/s corresponds approximately to Beaufort scale 5, at which purse-seine operations become difficult for artisanal vessels. This threshold was selected based on AIC comparison across 8, 10, and 12 m/s cut-offs on the trip-equation fit.

quotas, and environmental conditions into expected fishing effort under projected climate scenarios.

3.4 Projection approach

We assess the impact of climate change on the Chilean small pelagic fishery through two channels: (i) a *direct weather channel*, in which projected changes in wind speed alter the number of bad-weather days that constrain fishing operations and thus enter the negative binomial trip equation; and (ii) an *indirect biomass channel*, in which projected changes in SST and CHL modify each stock's intrinsic productivity $r_{i,t}$ through the climate shifter $(\rho_i^{SST}, \rho_i^{CHL})$ of Eq. (2) and, by that route, the law of motion in Eq. (1).

Climate projections are derived from a six-model CMIP6 ensemble—IPSL-CM6A-LR, GFDL-ESM4, CESM2, CNRM-ESM2-1, UKESM1-0-LL, and MPI-ESM1-2-HR—under two Shared Socioeconomic Pathways: SSP2-4.5 (moderate emissions) and SSP5-8.5 (high emissions). The ensemble was selected to span the CMIP6 distribution of equilibrium climate sensitivity. We apply the delta method (Burke et al., 2015; Free et al., 2019) independently to each model, which preserves the observed interannual variability and fine-scale spatial structure from satellite records while imposing the climate-change signal from CMIP6. For each model and variable, the delta is computed as the difference (additive for SST and wind speed) or, for chlorophyll-a, the difference of natural logarithms (so that the resulting $\Delta \log \mathrm{CHL}$ enters the shifter on its native estimation scale), between the CMIP6 future climatology and a spliced historical baseline that combines the CMIP6 historical run for 2000–2014 with the SSP2-4.5 run for 2015–2024 (CMIP6 historical runs end in 2014, so the splice is required to match the 2000–2024 estimation window of the structural model). For CESM2, which lacks chlorophyll under SSP2-4.5 in the NCAR CMIP6 catalogue, we splice the historical 2000–2014 segment with the SSP5-8.5 2015–2024 segment for chlorophyll only; the two scenarios are virtually indistinguishable in the near term and the substitution preserves CESM2's contribution to SSP5-8.5 chlorophyll deltas. We consider two future time windows: mid-century (2041–2060) and end-of-century (2081–2100). Comparative statics are then reported integrating over both the structural-parameter posterior and the cross-model dimension of the ensemble; the variance decomposition of the resulting projection uncertainty is reported in Appendix F for stock-level intrinsic productivity and in Appendix G for fleet-level trips.

For the direct weather channel, monthly wind speed deltas from CMIP6 are applied additively to the daily wind speed observations at each vessel's center of gravity (COG) over 2013–2024, and bad-weather days (wind speed exceeding 8 m/s) are recounted vessel by vessel under each scenario; the resulting $\Delta\mathrm{days}_{bw}$ enters the NB trip equation through the estimated semi-elasticity β_{weather}, retaining vessel-level heterogeneity in baseline weather exposure. Although the cross-model median CMIP6 wind anomaly is modest (below +0.5 m/s under SSP5-8.5 end-of-century), the historical daily wind distribution at the COG sits sufficiently close to the 8 m/s threshold that the cross-vessel median $\Delta\mathrm{days}_{bw}$ rises by +10 days under SSP2-4.5 mid-century and +23.5 days under SSP5-8.5 end-of-century. Note that uas and vas are missing from the CESM2 archive

in the CMIP6 DRS for this period, so the direct channel is computed from the five remaining ensemble members (CNRM-ESM2-1, GFDL-ESM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, UKESM1-0-LL) while the indirect biomass channel retains the full six-member ensemble where SST and chlorophyll deltas are available. For the indirect biomass channel, we evaluate the climate shifter of Eq. (2) at the posterior draws of $(\rho_i^{SST}, \rho_i^{CHL})$ from the full state-space specification, taking ΔSST and $\Delta \log CHL$ as the scenario- and window-specific CMIP6 anomalies relative to the 2000–2024 estimation baseline. This yields a full posterior distribution over $r_{i,t}^{\star}/r_i^0$ under each climate regime—a long-run comparative-statics object, in the sense that it describes the steady-state productivity of each stock under a counterfactual climate, not a year-by-year forecast of biomass. Forward simulation of biomass and catch trajectories under endogenous harvest rules is left to future work; the posterior reported here is archived to facilitate that extension.

As noted by Kasperski (2015), multi-species bioeconomic frameworks are well suited to long-horizon projections of fish populations rather than to intra-annual management responses. Because our empirical models are estimated on interannual variability rather than decadal trends, the projections reported here capture the mechanical effect of changed environmental conditions on stock productivity and trip counts while holding prices, quotas, and adaptive behaviour by fishers or managers fixed at historical levels (Auffhammer, 2018). They provide a useful benchmark for the direction and magnitude of climate impacts, with the understanding that behavioural responses may attenuate or amplify these effects over time.

4 Results

4.1 Identification of climate shifters

The central econometric object of this paper is the vector of stock-specific *climate shifters* $\{\rho_i^{SST}, \rho_i^{CHL}\}_{i=1}^3$, which modulate the intrinsic growth rate of each stock as

$$r_{i,t} \;=\; r_i^0 \,\exp\!\big(\,\rho_i^{SST}\,(SST_{t-1} - \overline{SST}) \;+\; \rho_i^{CHL}\,(\log CHL_{t-1} - \overline{\log CHL})\,\big).$$

These shifters admit a clean economic reading: ρ_i^{SST} is the semi-elasticity of the stock's intrinsic productivity r_i with respect to a sea-surface temperature anomaly of one degree Celsius, and similarly ρ_i^{CHL} is the semi-elasticity with respect to a log-point of chlorophyll-a anomaly (a common proxy for primary productivity at the base of the food web). They are the structural parameters that couple the biological law of motion $B_{i,t+1} = B_{i,t} + g_i(B_{i,t}, X_t) - C_{i,t} + \varepsilon_{i,t}$ to the physical climate, and are therefore the quantities that must be identified for the long-run projections of Section 4.3 to have meaning as *comparative statics under climate change* rather than as out-of-sample forecasts.

We adopt weakly informative priors on $\rho_i^{SST}, \rho_i^{CHL}$ derived from independent reduced-form stress

tests (Appendix A) and let the 2000–2024 biomass series update them. Table 1 reports the posterior means, 90% credible intervals, and the posterior-to-prior standard deviation ratio, which measures how much the data refine each parameter relative to the prior.

Table 1: Posterior identification of climate shifters.

Shifter	Stock	Prior $\mathcal{N}(\mu, \sigma)$	Posterior mean	Posterior 90% CI	$\sigma_{\text{post}}/\sigma_{\text{prior}}$	$\hat{R}$
ρ^{SST}	Anchoveta CS	$\mathcal{N}(-2.3, 1.0)$	-1.06	[-1.79, -0.38]	0.43	1.000
ρ^{SST}	Sardina comun CS	$\mathcal{N}(-2.0, 1.0)$	-2.75	[-3.46, -2.02]	0.44	1.000
ρ^{SST}	Jurel CS	$\mathcal{N}(0.0, 1.0)$	0.42	[-1.25, 2.09]	1.01	1.000
ρ^{CHL}	Anchoveta CS	$\mathcal{N}(-2.3, 1.0)$	-3.64	[-5.01, -2.25]	0.83	1.000
ρ^{CHL}	Sardina comun CS	$\mathcal{N}(2.1, 1.0)$	2.17	[0.96, 3.37]	0.74	1.000
ρ^{CHL}	Jurel CS	$\mathcal{N}(0.0, 1.0)$	-0.06	[-1.72, 1.60]	1.01	1.001

Note:
Shifters are the semi-elasticities of the intrinsic growth rate r_i with respect to SST anomalies (in °C) and log-CHL anomalies, respectively. Priors are adopted from the stock-specific stress-test results reported in Appendix A. The posterior-to-prior standard deviation ratio measures the amount of prior-to-posterior updating: values near one indicate the data carry little information about the parameter, whereas values well below one indicate identification from the likelihood. All R-hat <= 1.01.

The pattern of identification is itself informative and organises the economic interpretation of the paper. Three facts stand out.

First, for the two coastal upwelling stocks—anchoveta and sardina común—both climate shifters are identified with substantial precision. The posterior 90% credible intervals exclude zero for all four parameters, and the standard deviation ratios fall between 0.43 and 0.83, indicating that the 2000–2024 biomass series refines the priors materially. For anchoveta, a one-degree positive SST anomaly is associated with a semi-elasticity of $\rho^{SST}_{\text{anch}} = -1.06$ (CI $[-1.79, -0.38]$); for sardina común, the corresponding semi-elasticity is more than twice as large, $\rho^{SST}_{\text{sard}} = -2.75$ (CI $[-3.46, -2.02]$). Evaluated at one historical standard deviation of the SST anomaly series (estimation-sample $\hat{\sigma}_{SST} = 0.26\,^{\circ}\text{C}$ over 2000–2024), a thermal shock of that magnitude lowers the intrinsic growth rate of anchoveta by approximately 24% ($1 - e^{-1.06 \cdot 0.26}$) and that of sardina by approximately 51% ($1 - e^{-2.75 \cdot 0.26}$) relative to the climatological mean. Under a policy-relevant warming signal of $+1\,^{\circ}\text{C}$, comparable to mid-century CMIP6 SSP2-4.5 projections for the Chilean coastal zone, the implied declines are 65% for anchoveta and 94% for sardina, although $+1\,^{\circ}\text{C}$ lies approximately four historical standard deviations outside the estimation window and these projections therefore rely on the log-linear extrapolation embedded in the shifter function. Either way, the magnitudes are economically first-order for a fishery whose Centro-Sur TAC is in the low hundred-thousand-tonne range. The narrow historical SST variance ($\hat{\sigma}_{SST} = 0.26\,^{\circ}\text{C}$) also explains, mechanically, why the one-step-ahead forecasting test of Appendix B fails to discriminate between the climate-augmented and the climate-free specifications: within the estimation window the annual climate signal is small relative to the stochastic process innovation σ_{proc}, yet accumulates to economically consequential levels only over decadal horizons.

Second, the two coastal stocks respond to chlorophyll-a with **opposite signs**: $\rho^{CHL}_{\text{anch}} = -3.64$ (CI $[-5.01, -2.25]$) versus $\rho^{CHL}_{\text{sard}} = +2.17$ (CI $[+0.96, +3.37]$). At one historical standard deviation of $\log CHL$ ($\hat{\sigma}_{\log CHL} = 0.095$, a primary-productivity anomaly of roughly 10%), a positive CHL shock lowers anchoveta growth by approximately 29% while raising sardina growth by approximately 23%. This is a reduced-form footprint of the well-documented trophic asymmetry between the two species, summarised here for the non-specialist reader. Chlorophyll-a is a satellite-observable proxy for phytoplankton biomass, the primary producers at the base of the marine food web. Pulses of high chlorophyll-a indicate periods of strong primary productivity, typically driven by coastal upwelling that brings nutrient-rich deep water to the surface. Sardina común feeds directly on small phytoplankton and benefits from these pulses, so its productivity rises with chlorophyll-a anomalies. Anchoveta, by contrast, feeds on larger zooplankton whose populations are themselves preyed upon by small phytoplankton-feeders; when the food web is dominated by small primary producers, the zooplankton biomass that anchoveta relies on is suppressed, and anchoveta productivity falls. The opposite-signed semi-elasticities of ρ^{CHL}_{anch} and ρ^{CHL}_{sard} identified in Table 1 are consistent with this trophic asymmetry and are not a statistical artefact. The economic implication is first-order for this paper: the artisanal fleet, which lands both species, is not simply "climate-exposed"; it is exposed to **within-target heterogeneity** in the sign of climate effects. A warming-and-greening regime therefore does not translate into a uniform shock on the artisanal sector, but into a substitution across species that the current quota-allocation rules, indexed by species, cannot fully absorb.

Third, and in marked contrast, **neither SST nor CHL shifters are identified for jack mackerel**. The posterior standard deviations σ_{post} are indistinguishable from the prior scale ($\sigma_{\text{post}}/\sigma_{\text{prior}} \approx 1$), the posterior means lie near zero, and the credible intervals straddle zero in both directions. This is the expected result under the null hypothesis that jack mackerel productivity at the Centro-Sur scale is not locally coupled to the Chilean coastal upwelling regime—which is biologically sensible given its transboundary Southeast Pacific stock structure managed through SPRFMO. Econometrically, the model correctly reports non-identification rather than producing a spurious coefficient, which is a strength of the Bayesian state-space approach with structurally informed priors: the data can either move the prior or, as here, confirm that they are silent about it. This is the econometric dividend of Cowles-style structural identification in fishery climate-impact assessment: the structural posterior reports the absence of local climate signal where a reduced-form benchmark would have produced an artefactually precise but structurally meaningless coefficient, which would then propagate into climate projections as if it were information.

The non-identification result for jack mackerel is robust to the choice of spatial domain over which the environmental anomalies are constructed. Appendix E reports refits of the full state-space specification using SST and log-CHL averaged over two alternative dominios — an offshore-extended band (32°S–41°S, to 85°W) and a Southeast Pacific regional domain (20°S–45°S, to 90°W) — and confirms that the posterior-to-prior standard-deviation ratio of $(\rho^{SST}_{\text{jurel}}, \rho^{CHL}_{\text{jurel}})$ remains close to unity under all spatial aggregations considered. This supports the interpretation that the relevant climate forcing for the Centro-Sur jack mackerel stock operates at a scale not captured by Chilean

coastal anomalies, consistent with its transboundary SPRFMO management.

Figure 3 visualises the posterior–prior updating as a forest plot.

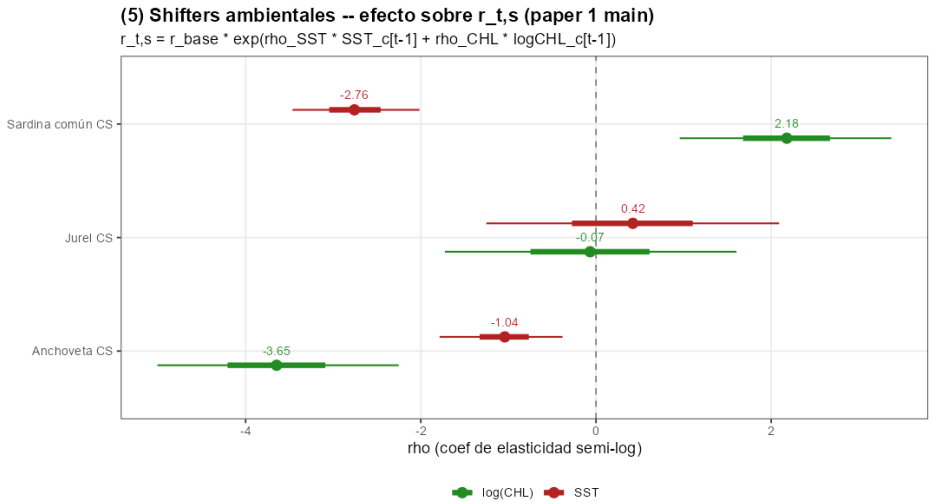

Figure 3: Posterior and prior distributions of the climate shifters $\rho_i^{SST}, \rho_i^{CHL}$ by stock. Anchoveta and sardina común show substantial posterior updating with narrow credible intervals excluding zero; jack mackerel posteriors coincide with the weakly informative prior centred at zero.

4.1.1 Identification versus short-run prediction

The identified shifters are structural parameters rather than reduced-form correlations: they measure an invariant biological mechanism linking ocean state to stock productivity, and they enter the law of motion as coefficients on exogenous climate inputs. This structural reading is what licenses their use as projection inputs under climate regimes outside the 2000–2024 estimation sample; reduced-form correlations, by contrast, would require the joint distribution of (SST, B, C) to remain stationary into the projection horizon, which is precisely the assumption the climate scenarios are designed to violate. Their economic value lies in **long-run comparative statics**: given a projected climate regime $X_t^\star$ for the period 2040–2100 drawn from the CMIP6 ensemble, the intrinsic productivity under that regime is $r_i^\star = r_i^0 \exp(\rho_i^{SST} \Delta SST_t + \rho_i^{CHL} \Delta \log CHL_t)$, and the expected steady-state biomass, harvest, and fleet-level effort follow mechanically. This is the channel through which the paper translates physical climate change into distributional outcomes in Section 4.3.

Crucially, identification of ρ does **not** imply superior short-run predictive power. A leave-future-out cross-validation at the one-year horizon (Appendix B) shows that the fully climate-augmented specification improves one-step-ahead log predictive density by only $\Delta\widehat{\mathrm{ELPD}}_{\mathrm{LFO}} \approx 0.4$ over a climate-free Schaefer baseline, a statistically unresolved difference. This is not a failure; it is precisely what should occur when the process-noise variance σ_{proc} is large relative to year-to-year

climate innovations. Annual climate variability contributes little to *next-year nowcasts* but accumulates systematically over decadal horizons, and the shifters ρ_i are the parameters that quantify that accumulation. A robustness comparison using PSIS-LOO is reported in Appendix B (Table 8) and confirms that the full specification is preferred as an in-sample description of the dynamics ($\Delta\widehat{\mathrm{ELPD}}_{\mathrm{LOO}} = 14.59$, SE $= 4.93$); the LOO and LFO results are complementary diagnostics of, respectively, identification-through-fit and forecasting-on-new-data, and the paper relies on the former for the economic interpretation.

4.1.2 Posterior-predictive adequacy

A stock-by-stock comparison of the smoothed latent biomass $B_{i,t}$ against the observed SSB series (Figure 8) shows that the identified model reproduces the historical trajectory within a median 90% posterior band width of approximately 20% of the mean. Residuals are roughly Gaussian and contain no detectable autocorrelation at one-year lag (Figure 9). Posterior-predictive diagnostics are reported in Appendix C; Markov-chain convergence diagnostics for all top-level parameters are reported in Appendix D.

4.2 Total annual trips

Table 2 reports the negative binomial estimates separately for each fleet segment. The specification includes year fixed effects, which absorb aggregate shocks affecting both fleets contemporaneously—most notably the 2019 social outbreak (*estallido social*) and the 2020–2022 COVID-19 period—and identify the structural coefficients β_H (allocated harvest) and β_{weather} (bad-weather days) from within-year cross-vessel variation. Without year fixed effects, $\hat{\beta}^{\text{ART}}_{\text{weather}}$ is approximately twice as large in magnitude, reflecting the contemporaneous correlation between high-wind years and the social/pandemic shocks; the year-FE specification is therefore the basis for the climate projections in Section 4.3 (the relative trip projection $T_v^{\text{FUT}}/T_v^{\text{HIST}}$ does not depend on the year-FE values themselves, which cancel in the ratio).

The coefficient on allocated harvest is positive and highly significant in both fleets, confirming that vessels with larger quota allocations undertake more trips per year. This variable provides the key linkage between the trip equation and the simulation framework: changes in regional TACs translate into vessel-level effort adjustments through the quota share mechanism.

The two fleets differ markedly in their price responsiveness. For the industrial fleet, the jack mackerel price is positive and significant ($p < 0.01$), consistent with this fleet's primary orientation toward jack mackerel. Sardine and anchoveta prices have no detectable effect on industrial effort, reflecting the limited participation of industrial vessels in the sardine-anchoveta fishery. For the artisanal fleet, the sardine price is positive and significant ($p < 0.01$), consistent with sardine being the dominant target species for artisanal vessels in the Centro-Sur region. The coefficient on the anchoveta price is negative and significant. This counterintuitive sign likely reflects simultaneity:

Table 2: Negative binomial estimates of annual fishing trips (2013–2024, $N_{IND} = 319$, $N_{ART} = 4201$)

	Dependent variable:	
	Industrial	Artisanal
	(1)	(2)
Hold capacity (log m^3)	0.005 (0.192)	0.221^{***} (0.052)
Allocated harvest (H_{vy}^{alloc}, tons)	0.00003^{***} (0.00001)	0.0005^{***} (0.00003)
Price jack mackerel (1000s \$/ton)	-0.018^{**} (0.008)	0.115^{***} (0.008)
Price sardine (1000s \$/ton)	-0.005 (0.005)	0.117^{***} (0.007)
Price anchoveta (1000s \$/ton)	-0.012^{**} (0.006)	0.159^{***} (0.007)
Bad weather days	-0.0001 (0.001)	-0.001^{*} (0.0005)
Veda days	-0.081^{***} (0.004)	0.013^{***} (0.002)
Vessel type (UNK)		-0.461^{***} (0.107)
Vessel type (BR)		-1.006^{***} (0.162)
Vessel type (BRV)		-2.395^{***} (0.289)
Vessel type (L)		-0.067 (0.137)
Vessel type (LM)	-1.318^{***} (0.167)	0.169 (0.134)
Constant	21.860^{***} (2.151)	-64.144^{***} (3.232)
Year fixed effects	Yes	Yes
Observations	319	4,201

Note: *p<0.1; **p<0.05; ***p<0.01.
Vessel-clustered robust SEs. Prices in constant 2018 thousands of pesos/ton.
Allocated harvest = historical vessel share × lagged aggregate harvest.
Year fixed effects absorb aggregate shocks (estallido social 2019, COVID 2020–2022).
LR test NB vs Poisson: Industrial $\chi^2 = 420$; Artisanal $\chi^2 = 40029$ (both $p < 0.001$).

years of low anchoveta availability produce both higher prices (through the inverse demand channel) and fewer trips (through reduced catch opportunities), generating a negative reduced-form correlation. As noted above, the purpose of this equation is not causal identification but rather to provide an empirically grounded mapping for the simulation, where prices are determined endogenously by the inverse demand module.

Hold capacity is not significant for the industrial fleet, where vessels are relatively homogeneous in scale, but is positive and significant for the artisanal fleet. Among artisanal vessels, larger hold capacity is associated with more annual trips, consistent with larger vessels being more commercially active and better able to sustain operations across varying conditions.

The environmental and regulatory variables show distinct patterns across fleets. For the artisanal fleet, adverse weather days carry a negative and statistically distinguishable coefficient, indicating that, controlling for year-specific aggregate conditions, vessels operating in grids with more days of wind speeds above 8 m/s undertake fewer fishing trips. For the industrial fleet, adverse weather is not significant, consistent with larger vessels being better equipped to operate under rough conditions. Biological closure days are strongly negative for the industrial fleet, representing the largest effect in the model, consistent with the industrial fleet's dependence on access during open seasons. For the artisanal fleet, the positive coefficient on closure days does not admit a causal interpretation: under the year-fixed-effects specification, β_{closure} is identified solely from the cross-zone differential between regions V-VIII (151 days) and IX-XIV (182 days) of the proxy variable described in Appendix A and discussed at length in the regulatory-history caveat of Section 5. The positive sign therefore reflects locational heterogeneity in artisanal fishing intensity between the two regulatory zones (the southern zone IX-XIV contains both more nominal closure days and more concentrated artisanal effort per vessel, and the cross-sectional comparison cannot disentangle the two), not the causal effect of closures on trips. Constructing a year-by-year closure variable with within-zone temporal variation that would identify the causal effect requires reconciling the nominal regulatory ceilings of the D.Ex. 115/1998 and 530/2016 chain with the actual realised closure days reported in the SUBPESCA Indicadores Biológicos portal from 2019 onwards, a non-trivial data-engineering task discussed in Section 5 and reserved for follow-up work.

Vessel type dummies capture residual heterogeneity in fleet composition. Among artisanal vessels, larger vessel categories (L, BRV) are associated with substantially fewer trips relative to the baseline, reflecting differences in trip duration and operational patterns across vessel classes.

4.3 Climate change projections

The projection methodology is summarized in Section 3.4. Under each scenario and time window we translate CMIP6-derived environmental deltas into (i) changes in bad-weather days entering the trip equation (direct channel) and (ii) posterior draws of the stock-specific effective growth rate $r^{\star}_{i,t}$ obtained by evaluating the identified climate shifter of Eq. (2) at the projected SST and log-CHL

anomalies (indirect channel).

4.3.1 Projected environmental changes

Table 3 summarizes the projected environmental changes for the Centro-Sur study area, reported as cross-model medians with the inter-model interquartile range (IQR) in brackets. Chlorophyll-a is reported as a percentage change, $\%\Delta\mathrm{CHL} = (\exp(\Delta \log \mathrm{CHL}) - 1) \times 100$, computed per model and then summarised across the ensemble; the structural model is fitted on $\Delta \log \mathrm{CHL}$ and that scale is used in all downstream calculations (Section 3.4). Sea-surface temperature warms by a cross-model median of +0.6 to +2.6 °C depending on the scenario, with the strongest warming under SSP5-8.5 end-of-century and an inter-model spread that grows with the forcing. The chlorophyll-a signal is small in magnitude relative to the cross-model spread: the cross-model median of $\%\Delta\mathrm{CHL}$ is close to zero in every scenario–window pair while individual models disagree on the sign, reflecting the well-documented inter-model disagreement on primary-productivity changes under climate forcing in CMIP6. Near-surface wind speed increases modestly under all scenarios (cross-model median +0.1 to +0.4 m/s), with a larger increase under SSP5-8.5 end-of-century than the single-model IPSL projection on which an earlier version of this paper was based; the inter-model IQR remains comparable to the median. While these mean wind changes are small, they translate into a substantial increase in the number of bad-weather days exceeding the 8 m/s operability threshold (cross-vessel median +10 to +23.5 days/year, see Section 3.4) because the historical daily wind distribution at vessel COGs sits sufficiently close to the threshold that a small upward shift in the mean displaces a non-negligible mass of the upper tail past it. The implications for fleet-level trips are quantified in Table 5 and are first-order for the artisanal fleet.

Table 3: Projected environmental changes for Centro-Sur Chile (CMIP6 six-model ensemble, delta method; cross-model median with inter-model IQR in brackets).

SSP	Window	Δ SST (°C)	Δ CHL (%)	Δ wind speed (m/s)
SSP2-4.5	2081–2100	+1.44 [+0.96, +1.91]	+0.0 [−3.6, +1.4]	+0.19 [+0.10, +0.24]
SSP2-4.5	2041–2060	+0.74 [+0.52, +0.92]	+2.2 [+0.0, +2.8]	+0.14 [+0.13, +0.15]
SSP5-8.5	2081–2100	+2.39 [+1.93, +3.29]	−1.3 [−7.4, +3.0]	+0.42 [+0.36, +0.45]
SSP5-8.5	2041–2060	+0.92 [+0.61, +1.23]	+0.7 [−2.3, +3.4]	+0.21 [+0.14, +0.32]

4.3.2 Comparative statics on intrinsic stock productivity

Table 4 reports the comparative-statics object $\Delta r_i^\star / r_i^0$ obtained by evaluating the identified climate shifter at the CMIP6 ensemble deltas of Table 3 and integrating over the posterior of $(\rho_i^{SST}, \rho_i^{CHL})$ under the full state-space specification. For each scenario–window pair we compute the within-model posterior of $\Delta r_i^\star / r_i^0$ for each CMIP6 model in turn and then summarise the resulting set of model-level distributions along two distinct axes: the cross-model median and inter-quartile range (IQR) of the model-level posterior medians, which captures inter-model climate

uncertainty, and the median across models of the within-model 90% posterior credible interval, which captures structural uncertainty about $(\rho_i^{SST}, \rho_i^{CHL})$ holding the climate forcing fixed. The two intervals are reported separately because they have different policy implications and a separate variance decomposition is reported in Appendix F. Jack mackerel is reported as non-identified ("n.i.'') because its shifter posterior coincides with the weakly informative prior (Section 4.1), so that projecting its productivity under future climate regimes would merely propagate prior uncertainty rather than transmit evidence from the 2000–2024 sample. A forward bioeconomic simulation for jack mackerel, combining the Centro-Sur stock with the SPRFMO transboundary assessment, is left to future work.

Table 4: Comparative statics on intrinsic stock productivity under the CMIP6 six-model ensemble.

Stock	Scenario	n_m	ΔSST (°C)	$\Delta \log$ CHL	$\Delta r_i^\star / r_i^0$ (cross median)	Cross-model IQR	Within posterior 90% CI	Pr($\Delta < 0$)
Anchoveta CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	-50.7%	[-52.7%, -47.8%]	[-68.0%, -27.0%]	1.00
Anchoveta CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	-71.0%	[-76.3%, -60.4%]	[-91.1%, -31.0%]	1.00
Anchoveta CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	-62.4%	[-68.0%, -55.9%]	[-80.9%, -31.2%]	1.00
Anchoveta CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	-89.6%	[-91.5%, -87.4%]	[-98.3%, -54.2%]	0.99
Sardine CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	-78.6%	[-93.2%, -72.5%]	[-85.7%, -67.3%]	1.00
Sardine CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	-97.2%	[-98.9%, -89.8%]	[-98.9%, -92.6%]	1.00
Sardine CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	-91.1%	[-96.7%, -78.5%]	[-95.1%, -83.2%]	1.00
Sardine CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	-99.9%	[-100.0%, -99.4%]	[-100.0%, -99.3%]	1.00
Jack mackerel CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	n.i.	n.i.	n.i.	n.i.

Note:
Entries summarise the posterior of the proportional change in intrinsic productivity, Delta r / r0 = exp(rho_SST * Delta SST + rho_CHL * Delta log CHL) - 1, evaluated at scenario-window-specific CMIP6 anomalies for each of n_m models in the ensemble and integrated over the posterior of (rho_SST, rho_CHL) from the full state-space specification. The cross-model IQR is the 25th-to-75th percentile range across CMIP6 models of the within-model posterior medians and captures inter-model climate uncertainty. The within posterior 90% CI is the cross-model median of the within-model 5th-to-95th-percentile posterior interval and captures structural uncertainty about (rho_SST, rho_CHL) holding the climate forcing fixed. Jurel (jack mackerel) CS is reported as non-identified ('n.i.') because the posterior of its shifters coincides with the weakly informative prior; see the identification discussion in Results. n_m = 5 instead of 6 in the SSP2-4.5 panels because CESM2 does not publish chl-ssp245 in the NCAR CMIP6 catalogue. Under SSP5-8.5 end-of-century the implied SST anomaly lies approximately nine in-sample standard deviations outside the realised 2000-2024 envelope, so it is best read as a structural-form stress test under the maintained log-linear shifter. Raw numerical values are archived in tables/growth_comparative_statics_raw.csv; model-level disaggregation is in tables/growth_comparative_statics_by_model.csv. Appendix F reports the variance decomposition between cross-model and within-model components.

The magnitudes are economically first-order for a Centro-Sur fishery whose combined TAC is in the low hundred-thousand-tonne range. Under the moderate SSP2-4.5 mid-century scenario (cross-model median +0.58 °C; $\Delta \log$ CHL near zero in the cross-model median, with non-trivial inter-model spread), the cross-model median of anchoveta's intrinsic productivity declines by 51% and sardina común's by 79%, with posterior probability of negative impact equal to one in both cases. Under the high-emissions SSP5-8.5 end-of-century scenario (cross-model median +2.39 °C, $\Delta \log$ CHL cross-model median -0.013 with 5th-percentile of -0.30 across models) the cross-model medians deepen to -90% and -99.9% respectively, and the entire 90% credible interval of sardine's response lies below zero in every model of the ensemble; the anchoveta cross-model probability of decline is 0.99 rather than 1 because under a small fraction of (model, draw) pairs a sufficiently negative $\Delta \log$ CHL combined with $\rho_{\text{anch}}^{CHL} < 0$ offsets the SST-driven collapse. The identified signs confirm the qualitative reading of the shifters in Section 4.1: both coastal stocks respond negatively to SST, but sardina común carries roughly twice the SST semi-elasticity of anchoveta, and its CHL semi-elasticity is of the opposite sign, so that the compound response—negative thermal shock combined with negative primary-productivity shock—is sharper for sardina than for anchoveta.

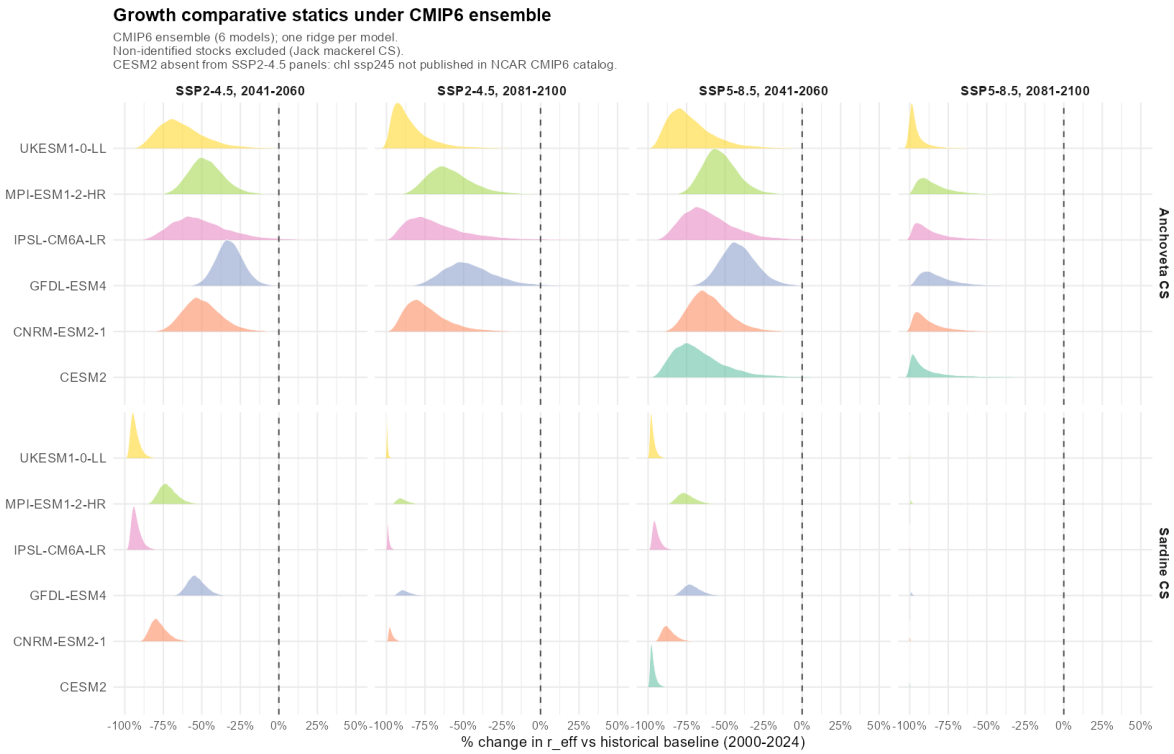

Figure 4: Posterior distribution of the proportional change in intrinsic stock productivity under CMIP6 scenarios, six-model ensemble. Ridges are kernel-smoothed marginal posteriors of the proportional change in intrinsic productivity derived from draws of the state-space full specification, with one ridge per CMIP6 model in each scenario–window pair; vertical line at zero. CESM2 is absent from the SSP2-4.5 panels because chl-ssp245 is not published in the NCAR CMIP6 catalogue. Jack mackerel is omitted because its climate shifter is not identified in the 2000–2024 sample.

The comparative-statics exercise is a long-run object rather than a year-by-year forecast. It describes what the steady-state intrinsic productivity of each stock would be under a counterfactual climate, *holding fixed* the law of motion and the fishery's regulatory regime. This is precisely what the structural identification of the shifters permits: because $(\rho_i^{SST}, \rho_i^{CHL})$ are invariant to the joint distribution of (SST, B, C), they can be evaluated at climate regimes that lie outside the estimation window—in particular at an SST anomaly of $+2.3^{\circ}$C that is roughly nine historical standard deviations beyond the 2000–2024 mean. Reduced-form correlational specifications do not license this extrapolation.

4.3.3 Implications for fleet-level effort

Table 5 translates the comparative-statics object on intrinsic stock productivity reported in Table 4 into a comparative-statics object on fleet-level annual trips, by propagating each posterior draw through the Schaefer steady-state biomass equation under historical average fishing pressure and through the negative binomial trip equation of Section 4.2. Vessel-level responses are aggregated

to the fleet level as medians over posterior draws and vessels within fleet; the aggregation weights are each vessel's realized catch composition over the 2012–2024 estimation window.

Table 5: Long-run implications of CMIP6 climate scenarios for fleet-level fishing effort, under a Schaefer steady-state thought experiment with fishing pressure fixed at the 2000–2024 historical average. Entries are cross-model medians with cross-model IQRs in brackets, aggregated within each CMIP6 model over posterior draws and vessels within fleet.

Fleet	Scenario	$\Pr(f_v^H < 0.5)$	%Δ trips (marginal)	%Δ trips (conditional)
Artisanal	SSP2-4.5, 2041-2060	0.95 [0.93, 0.96]	-8.1% [-8.3%, -7.7%]	-2.3% [-2.9%, -2.3%]
Artisanal	SSP2-4.5, 2081-2100	0.98 [0.97, 0.99]	-9.3% [-9.3%, -8.7%]	-2.1% [-2.3%, -2.0%]
Artisanal	SSP5-8.5, 2041-2060	0.98 [0.96, 0.98]	-8.7% [-8.9%, -8.3%]	-2.2% [-2.8%, -1.9%]
Artisanal	SSP5-8.5, 2081-2100	0.99 [0.99, 0.99]	-10.2% [-10.5%, -10.0%]	-2.7% [-2.9%, -2.3%]
Industrial	SSP2-4.5, 2041-2060	0.12 [0.11, 0.12]	-0.7% [-0.7%, -0.7%]	-0.6% [-0.6%, -0.6%]
Industrial	SSP2-4.5, 2081-2100	0.12 [0.12, 0.12]	-0.8% [-0.8%, -0.7%]	-0.7% [-0.7%, -0.6%]
Industrial	SSP5-8.5, 2041-2060	0.12 [0.12, 0.12]	-0.7% [-0.8%, -0.7%]	-0.6% [-0.7%, -0.6%]
Industrial	SSP5-8.5, 2081-2100	0.12 [0.12, 0.12]	-0.9% [-0.9%, -0.8%]	-0.8% [-0.8%, -0.7%]

Note:
Entries summarise the posterior of the proportional change in fleet-level annual trips, computed for each of the six CMIP6 models in the ensemble and then aggregated across models. Within each model, the posterior is integrated over draws of the full state-space specification and vessels within fleet (composition-weighted exposure). Cross-model medians and 25th-to-75th-percentile inter-quartile ranges across the n_m models are reported in brackets; the median across models of the within-model 5th-to-95th-percentile posterior interval (within posterior CI) is archived in tables/trip_comparative_statics_raw.csv. f_H is the composition-weighted biomass factor relative to the historical baseline: f_H = sum over species of omega_vs * B_star_s / B_hist_s, where B_star_s = K_s * (1 - F_hist_s / r_eff_s) is the Schaefer steady-state biomass and omega_vs is the vessel's realized catch share in species s. Pr(f_H < 0.5) reports the posterior probability that the vessel's composition-weighted biomass factor falls below one-half. The marginal median integrates over all posterior draws within each model; the conditional median restricts to draws with f_H >= 0.5, isolating the pure climate-elasticity component from the steady-state collapse component of the trip response. Jack mackerel is treated as non-identified (factor_B = 1) consistent with the identification discussion in the Results section. Raw numerical values, by-model disaggregation, and by-stock extinction probabilities are archived in tables/trip_comparative_statics_raw.csv, tables/trip_comparative_statics_by_model.csv, and tables/trip_comparative_statics_extinct.csv. The narrow cross-IQR observed in many cells reflects a floor effect: once Schaefer steady-state collapse saturates the portfolio-weighted biomass factor across models, the trip response converges to exp(-beta * H_alloc * 1), independent of climate magnitude. Appendix G reports the formal two-way variance decomposition of the fleet-level trip response, which confirms that the within-model component (posterior plus vessel heterogeneity) dominates the total variance under saturated regimes; Appendix F reports the analogous decomposition for the underlying productivity response.

The posterior probability of portfolio loss—defined as a reduction of more than fifty percent in the composition-weighted biomass factor f_v^H relative to the 2000–2024 historical baseline—exceeds 0.95 for the artisanal fleet under every CMIP6 scenario considered and reaches 0.99 under SSP5-8.5 end-of-century. The industrial fleet exhibits a loss probability of 0.12 that is stable across all four scenarios, driven entirely by its five-percent exposure to sardina común; the remaining 95% of its historical portfolio is allocated to jack mackerel, whose climate shifter is treated as non-identified in Section 4.1. The asymmetry in loss probability between the two fleets is first-order and stable across climate pathways, approximately eight-fold across all four scenarios.

Decomposing the fleet-level trip response isolates the channels through which this asymmetry

operates. The *marginal* posterior median of $\%\Delta$ trips, computed over all posterior draws, ranges from -8.1% (SSP2-4.5 mid) to -10.2% (SSP5-8.5 end) for the artisanal fleet and from -0.7% to -0.9% for the industrial fleet across the same scenarios—a ratio of roughly eleven to one at end-of-century. The *conditional* posterior median, computed over the subset of posterior draws in which $f_v^H \geq 0.5$, isolates the climate-elasticity component from the portfolio-collapse component and ranges from -2.1% to -2.7% for the artisanal fleet and from -0.6% to -0.8% for the industrial fleet, a conditional ratio of roughly three-and-a-half to one. Two complementary mechanisms generate this asymmetry. First, the indirect biomass channel transmits the climate signal asymmetrically through differences in portfolio composition: the artisanal fleet is concentrated in the two coastal-upwelling stocks whose climate shifters $(\rho_i^{SST}, \rho_i^{CHL})$ are both identified and economically large, while 95% of the industrial fleet's historical portfolio sits in jack mackerel, whose Centro-Sur shifter is not identified and is therefore held at the climatological mean in the projection. Second, the direct weather channel transmits the climate signal asymmetrically through differences in fleet-level weather sensitivity: the negative binomial trip equation estimates $\widehat{\beta}_{\text{weather}}^{\text{ART}} = -0.0008$ (statistically distinguishable from zero) and $\widehat{\beta}_{\text{weather}}^{\text{IND}} = -0.0001$ (statistically indistinguishable from zero), so the projected $+10$ to $+23.5$ days of additional bad weather per year translate into a -0.7% to -1.9% contribution to the artisanal trip response and a contribution close to zero for the industrial fleet, consistent with the size-dependent weather tolerance of the two segments. Both $\widehat{\beta}_{\text{weather}}$ estimates are obtained from the negative binomial specification with year fixed effects (Section 4.2), which absorbs the aggregate shocks of the 2019 social outbreak and the 2020–2022 COVID period; without year fixed effects, $\widehat{\beta}_{\text{weather}}^{\text{ART}}$ is approximately twice as large in magnitude, indicating that the contemporaneous correlation between high-wind years and the social/pandemic shocks would otherwise inflate the estimated weather elasticity. The LMCA's limited cross-sector transferability locks this dual exposure—portfolio composition under the indirect channel and weather sensitivity under the direct channel—into differential losses of harvest capacity, echoing the portfolio mechanism documented for U.S. West Coast and Alaskan fisheries by Kasperski & Holland (2013) and the regime-shift buffering documented by Cline et al. (2017): segments with narrower species portfolios face systematically higher exposure to species-specific shocks, climate-driven or otherwise, and institutional access constraints of the type imposed by the LMCA can limit the realized benefits of diversification (Oken et al., 2021).

Three caveats qualify this thought experiment. First, the Schaefer steady-state comparative statics holds fishing pressure fixed at the 2000–2024 historical average; any regulatory response that lowered F in the face of falling productivity would attenuate both the loss probability and the marginal trip response reported here. Quantifying that regulatory margin requires a forward simulation of the biomass process under endogenous harvest rules—most prominently the LTP cap on industrial quota transferability and the annual SUBPESCA revision of TACs in response to in-year biomass signals—which we leave to future work and which would reuse the posterior reported here. Second, the 2000–2024 observation window was not itself in Schaefer equilibrium under F_{hist}; the pipeline's internal consistency check, obtained by evaluating the comparative statics at $\Delta SST = 0$

and $\Delta \log CHL = 0$, yields a median f_v^H of approximately 1.015 rather than 1.000, reflecting a mild historical depletion from equilibrium. The scenario results in Table 5 should therefore be read as responses *relative to the baseline steady-state* rather than relative to the 2000–2024 observed trajectory. Third, the jack mackerel shifter is treated as non-identified, a finding that is empirically robust across three complementary tests reported in Appendix E. Re-fitting the state-space on three nested spatial domains for the environmental aggregation (Centro-Sur EEZ, offshore-extended, regional Southeast Pacific) yields posterior-to-prior standard deviation ratios of 0.998–1.014 for both $\rho^{SST}_{\text{jurel}}$ and $\rho^{CHL}_{\text{jurel}}$, indistinguishable from unity in every domain (Section E.3). A dual-source extension that augments the likelihood with the Northern Chilean acoustic series under a common climate elasticity—an assumption motivated by the 0.88 log-scale correlation between the Centro-Sur and Northern series across overlapping years—reduces the ratios only to 0.94–0.99, well above the threshold for material identification (Section E.4). A separate exploratory attempt to incorporate the SPRFMO range-wide assessment (OROP-PS, encompassing Chile, Peru, Ecuador and the adjacent high-seas area) fails on coherence grounds, with multimodal posterior support and unsatisfactory chain mixing, indicating that the broader geographic aggregate is not integrable with the Centro-Sur record under a single shifter. The convergence of these tests indicates a genuinely flat shifter within the Centro-Sur biological domain, not an artefact of insufficient spatial extent, insufficient sample size, or poor source aggregation. Two mechanisms remain observationally equivalent and cannot be distinguished within the available record: either jack mackerel productivity at the Centro-Sur scale is genuinely decoupled from Chilean coastal-upwelling forcing, as biological theory predicts for a highly migratory transboundary species managed at the SPRFMO scale, or the relevant forcing operates at a spatial scale outside the Centro-Sur biological domain to which the available data give us empirical access. The industrial fleet's apparent climate protection in Table 5 therefore rests on a null that cannot be sharpened within the Centro-Sur identification strategy, even with the dual-source extension; the appropriate setting in which to discriminate between the two mechanisms is a range-wide SPRFMO analysis with explicit cross-stock spillover, which we also leave to future work.

5 Discussion

Our results reveal a substantial asymmetry in how climate change affects different segments of Chile's small pelagic fishery, governed by the differential exposure of each fleet's species portfolio to the identified climate parameters under the LMCA's limited cross-sector transferability. The two coastal upwelling stocks targeted by the artisanal fleet—anchoveta and sardina común—both respond negatively to warming, with sardina común carrying roughly twice the SST semi-elasticity of anchoveta and an opposite-signed CHL response; the industrial fleet is diversified across these two coastal stocks and the transboundary jack mackerel stock, whose local productivity is *not* identified in the 2000–2024 sample. The artisanal fleet, concentrated in the sardina–anchoveta pair, is therefore the segment whose long-run harvest capacity is most exposed to the CMIP6 climate

signal under the current LMCA architecture. This finding aligns with the broader literature on heterogeneous climate impacts within fleets (Free et al., 2019; Sumaila et al., 2011) and underscores that aggregate projections can mask first-order distributional consequences inside the same fishery.

The fact that the jack mackerel shifter is not identified is itself a substantive result rather than a nuisance. It is what the Bayesian state-space specification correctly reports when a stock's Centro-Sur productivity is decoupled from local coastal forcing, as biological theory would predict for a highly migratory, transboundary species managed at the SPRFMO scale (Arcos et al., 2001). Reduced-form specifications that force a coefficient onto this species would instead deliver a precisely estimated but structurally meaningless number, which would then be propagated into climate projections as if it were information. The state-space identification exposes the absence of information directly; this is the econometric value-added of Cowles-style structural inference in this setting, as opposed to correlational benchmarks. The corollary for policy is that climate adaptation for jack mackerel cannot be read off the Chilean Centro-Sur time series alone and must be assessed jointly with the SPRFMO stock assessment in future work.

The two channels through which climate change reaches fleet-level effort operate on very different scales and on different segments of the fleet. The direct weather channel is small in mean wind anomaly—the cross-model median CMIP6 increase in near-surface wind speed is below +0.5 m/s even under SSP5-8.5 end-of-century—but the historical daily wind distribution at vessel COGs sits sufficiently close to the 8 m/s operability threshold that the additional bad-weather days exceed twenty per year on average under SSP5-8.5 end-of-century. This translates into a -0.7% to -1.9% marginal contribution to the artisanal trip response and a contribution indistinguishable from zero for the industrial fleet, reflecting the differential weather sensitivity of the two segments captured by $\hat{\beta}_{\text{weather}}$. The indirect biomass channel dominates the response in absolute terms for both fleets but operates entirely through portfolio composition, so the asymmetry it generates is intrinsic to the LMCA's allocation of stocks across segments. Climate adaptation policy in the Centro-Sur therefore needs to address both margins: the indirect channel through reform of the institutional architecture that mediates portfolio exposure—quota allocation rules, transferability across fleets, and the design of TACs under shifting species productivity—and the direct channel through operational measures targeted at the artisanal segment specifically, such as expanded port infrastructure, vessel reinforcement programs, or weather-indexed insurance, which are first-order for that fleet but redundant for the industrial purse-seine fleet. This conclusion uses the CMIP6 monthly wind ensemble; downscaled regional climate models that resolve extreme wind events could only raise the weight of the direct channel further.

The species-level productivity projections deserve careful interpretation. The log-linear form of the climate shifter in Eq. (2) is identified from interannual variation over 2000–2024, during which the realised range of SST anomalies was narrow ($\hat{\sigma}_{SST} \approx 0.26\,^{\circ}\text{C}$). CMIP6 anomalies under SSP5-8.5 end-of-century reach roughly $+2.3\,^{\circ}\text{C}$, which lies nearly an order of magnitude outside the estimation window; the corresponding posterior of $\Delta r_i^{\star}/r_i^0$ therefore leans heavily on the log-linear

extrapolation embedded in the shifter function. We regard the projected magnitudes as indicative of the direction and relative scale of impacts across stocks rather than as precise point forecasts. The log-linear shifter is a maintained simplification: thermal-tolerance arguments in the climate-fisheries literature (Cheung et al., 2010; Free et al., 2019) imply that for stocks operating below their thermal optimum the implied productivity response under high warming is, if anything, *understated* by a log-linear specification, so the projected magnitudes for SSP5-8.5 end-of-century should be read as conservative within the class of monotone shifter functions consistent with the identified posterior signs. A non-parametric extension of the shifter, or a pooled identification across multiple Humboldt-Current datasets, is the natural direction in which to relax this assumption.

Several caveats apply to the present analysis. First, the projections hold ex-vessel prices and management rules constant at historical levels, implying that the long-run responses capture the mechanical effect of changed environmental and biological conditions without accounting for endogenous price adjustments or adaptive management. Sumaila et al. (2011) note that reduced fish supply under climate change could increase prices, partially offsetting revenue losses from lower catches, while Lam et al. (2016) project sizable changes in global fisheries revenues by the 2050s once price responses and cross-country trade are accounted for. Incorporating an inverse demand system and numerical optimization of quota paths is a natural extension to capture these equilibrium adjustments. Second, although our projections integrate over a six-model CMIP6 ensemble, the cross-model spread in chlorophyll-a remains substantial: the cross-model median of $\Delta \log \mathrm{CHL}$ is close to zero in all scenario–window pairs while the inter-model dispersion is comparable in magnitude to the median, reflecting the well-documented disagreement among Earth system models on the direction and magnitude of primary productivity changes (Sumaila et al., 2011). The variance decomposition reported in Appendix F shows that, for the sardine projection, the between-model component accounts for 77–91% of total projection variance under three of the four scenario–window pairs, so the residual ensemble uncertainty on this stock is large despite the multi-model integration. The fleet-level trip projections in Table 5, by contrast, exhibit narrow cross-model interquartile ranges that should not be read as climate consensus: the corresponding decomposition reported in Appendix G attributes 97–100% of the total variance in $\Delta T_f/T_f^{(0)}$ to the within-model component (posterior pooled with vessel heterogeneity within fleet), reflecting a floor-effect saturation in which the Schaefer steady-state biomass collapse drives the trip response into a fleet-specific plateau independent of the climate magnitude. The substantive implication is that the marginal information value of expanding the CMIP6 ensemble for fleet-level trip projections is small relative to that of tightening the structural posterior on $(\rho_s^{SST}, \rho_s^{CHL})$ or refining the description of vessel heterogeneity in $(\omega_{v,s}, H_v^{\mathrm{alloc}})$; what governs the heterogeneity in trip outcomes across vessels and fleets is not which CMIP6 model is correct—given that all of them imply portfolio collapse for the two coastal stocks—but the differential economic exposure to that collapse imposed by historical catch composition and quota allocation. Third, the shifters are identified on a 25-year time series and, while the state-space specification propagates process-noise and measurement-noise uncertainty into the posterior, its ability to detect non-linearities in the climate response is limited.

The posterior-to-prior standard-deviation ratios reported in Table 1 provide a direct empirical check that the data update the prior for anchoveta and sardina común (ratios in the 0.43–0.83 range), while leaving the jurel shifters at the prior (ratios essentially equal to one)—the latter being the basis of our non-identification finding. Fourth, our models do not explicitly incorporate risk preferences or production risk, which may play a role in shaping fleet-level responses to environmental variability (Kasperski & Holland, 2013; Sethi et al., 2014). Fifth, the biological-closure variable in the trip equation is coded as a zonal proxy with no within-year temporal variation, set to the nominal regulatory ceiling of the reproductive closure regime under the relevant SUBPESCA decrees: the V-VIII zone receives 151 days and the IX-XIV zone receives 182 days throughout the panel. The underlying regulatory chronology is in fact non-trivial—D.Ex. 115/1998 governed the closure regime through July 2016 with annual modifications (D.Ex. 1661/2009 establishing 21 August to 21 October as the standing period, with year-specific adjustments in D.Ex. 705/2010, 796/2012, 747/2013, and 598/2015); D.Ex. 530/2016 then introduced a referential period of 6 July to 31 October with a fixed core (3 August to 4 October) and dynamic windows triggered by weekly IGS/PHA biological indicators; D.Ex. 137/2021 refined the indicator thresholds; and D.Ex. 05/2024 superseded the 530/2016 decree—and the actual realised closure days reported by the SUBPESCA Indicadores Biológicos portal range from 98 (in 2019) to 230 (in 2024) across regions and years. Constructing a year-by-year closure variable that reconciles the nominal regulatory regime with the actual realised closure days reported in the portal (available from 2019 onward) and with auxiliary recruitment-closure decrees not regulated by the reproductive-closure chain documented above is a non-trivial data-engineering task that we leave to follow-up work; for the present paper the zonal proxy captures the structural V-VIII / IX-XIV differential (31 days) that, under the year-fixed-effects specification of Section 4.2, is the variation that identifies β_{closure}. Sixth, the trip equation panel is constructed from IFOP logbook records and is therefore representative of the purse-seine fleet that operates under the LMCA quota architecture; smaller artisanal vessels using non-purse-seine gears (lampara, lines) operate outside the logbook system. Cross-validation against the SERNAPESCA vessel-level landings database confirms that the panel's portfolio weights $\omega_{v,s}$ match the SERNAPESCA aggregate within two percentage points for the industrial fleet and within three percentage points (catch-weighted) for the purse-seine artisanal subset that the panel represents. The non-purse-seine artisanal segment, which accounts for approximately 30% of total Centro-Sur artisanal landings and exhibits a higher proportional exposure to jack mackerel, is not modeled here; a sample-weighting extension that incorporates that segment is left to future work. The biomass likelihood, by contrast, is constructed from SERNAPESCA all-gear landings and is unaffected by this restriction. Seventh, the direct weather channel is constructed from five of the six CMIP6 ensemble members because CESM2 does not publish near-surface wind components (uas, vas) in the CMIP6 DRS for the relevant scenarios and time windows; CESM2 contributes to the indirect biomass channel through SST and chlorophyll deltas (six members in SSP5-8.5, five in SSP2-4.5 by the chlorophyll restriction noted in Section 3.4) but enters the direct channel with $\Delta\text{wind} = 0$ by construction. The contribution of CESM2 to the cross-model spread of fleet-level

trips therefore flows entirely through the indirect channel, which is the dominant source of variance in any case under the floor-effect saturation documented in Appendix G.

The finding that climate change creates winners and losers within the same fishery has direct implications for quota allocation policy. Chile's current regime—the *Límite Máximo de Captura por Armador* (LMCA)—allocates quotas separately for the artisanal and industrial sectors and stipulates limited transferability across fleet segments. Under such a rule, the climate-driven divergence in stock productivity reported in Table 4 translates one-to-one into a divergent loss of effective quota value across the artisanal–industrial divide: this is a feature of the allocation rule, not of the climate signal alone. If the long-run productivity of the coastal-upwelling pair declines by the magnitudes reported in Table 4, the relative value of artisanal quotas—heavily concentrated in those stocks—could fall substantially, potentially motivating reforms to the LMCA to allow more flexible cross-sector transfers. This echoes the portfolio literature on fishery diversification (Cline et al., 2017; Kasperski & Holland, 2013), which shows that access to a broader species portfolio reduces revenue variability and buffers communities against abrupt regime shifts. Oken et al. (2021) further document that institutional access constraints — such as limits on transferability of fishing rights — can substantially reduce the realized benefits of diversification, which is precisely the channel through which the LMCA's cross-sector restrictions translate into amplified climate exposure for the artisanal segment. Secure, transferable rights across fleet segments would facilitate adaptation to climate-driven shifts in species composition (Holland & Kasperski, 2016).

6 Conclusions

This paper estimates a multi-species bioeconomic model for Chile's small pelagic fishery and quantifies the long-run productivity response of each stock to CMIP6 climate scenarios. We contribute to the literature in three ways. First, we identify—rather than merely estimate—a set of structural climate shifters $(\rho_i^{SST}, \rho_i^{CHL})$ that modulate the intrinsic growth rate of each stock in a Bayesian state-space specification calibrated on official IFOP and SPRFMO assessments, and we demonstrate that the resulting parameters are economically interpretable as semi-elasticities of stock productivity with respect to thermal and primary-productivity anomalies. Second, we evaluate these shifters under a six-model CMIP6 ensemble (IPSL-CM6A-LR, GFDL-ESM4, CESM2, CNRM-ESM2-1, UKESM1-0-LL, MPI-ESM1-2-HR) under SSP2-4.5 and SSP5-8.5, and report their comparative-statics implications for intrinsic productivity together with the full posterior of the response, separating cross-model uncertainty from within-model posterior uncertainty via the variance decomposition reported in Appendix F. The companion decomposition for fleet-level trips in Appendix G shows that, once the Schaefer steady-state collapse saturates the portfolio for the two coastal stocks, the within-model component (posterior pooled with vessel heterogeneity) accounts for essentially the entire total variance, so the policy-relevant uncertainty in the trip projections lies on the within-model side rather than in disagreement across CMIP6 models. Third, we link

the identified structural response to the negative binomial trip equations estimated separately for the artisanal and industrial fleets, and we show that the direction of the distributional asymmetry across fleets is governed by the interaction between portfolio composition and the LMCA's limited cross-sector transferability, with the artisanal segment bearing the sharper long-run loss because its historical landings are concentrated in the two stocks whose climate semi-elasticities are identified and economically large. This finding extends the portfolio mechanism documented by Kasperski & Holland (2013) and Cline et al. (2017) to the climate-impact setting and to a quota regime — the LMCA — that constrains the cross-sector reallocation of fishing rights, in line with the access-constraint argument of Oken et al. (2021).

Our projections indicate that the long-run productivity of the coastal-upwelling pair—anchoveta and sardina común—is poised to decline sharply under all CMIP6 scenarios considered, with sardina común consistently more exposed than anchoveta and with the SSP5-8.5 end-of-century scenario essentially eliminating the intrinsic productivity of sardina común at the Centro-Sur scale. The Centro-Sur jack mackerel stock, by contrast, does not provide sufficient information to identify a local climate response in the 2000–2024 sample, consistent with its transboundary SPRFMO management. These results suggest that management policies focused on aggregate effort levels may miss important heterogeneity across fleets and species, and that climate adaptation in this fishery will need to address the uneven distribution of impacts through the quota-allocation architecture rather than through operational margins.

Several extensions to the present framework are natural. A first extension uses the posterior of $(r_i^0, K_i, \rho_i^{SST}, \rho_i^{CHL}, \Omega)$ identified here as a prior for a Stackelberg bi-level optimization, in which the regulator chooses the time path of TACs and the artisanal and industrial fleets allocate effort across stocks under the climate-shifted law of motion. That exercise would deliver the full forward biomass trajectories under CMIP6 anomaly paths with 90% credible bands, complementing the comparative-statics treatment of the present paper, and would follow the complete Kasperski (2015) approach with trip-level cost functions and an inverse demand system. We leave it to future work; the posterior reported here is archived in machine-readable form to facilitate that extension. The spatial dimension of effort allocation is also a natural extension, connecting the multi-species model to the location choice literature (e.g., Dupont, 1993; Hicks et al., 2020; Smith, 2005), since the geographic distribution of species availability is likely to shift under warming. A direct route for this spatial extension is to migrate the annual NB trip equation to a daily discrete choice specification with environmentally-informed species distribution models as proxies for local availability, following the approach of Quezada-Escalona et al. (2026) for the U.S. West Coast Coastal Pelagic Species fishery. SDMs for the Chilean Humboldt Current System are an active area of development in IFOP and academic collaborations, and once available at sufficient spatial resolution for anchoveta, sardina común, and jack mackerel, they would allow the present framework to capture the daily and within-port heterogeneity in fishing-effort responses to climate-driven shifts in availability. Finally, incorporating risk preferences and production risk—through, for example, random coefficient models—would provide a richer characterization of how fishers respond to the increased

environmental variability projected under climate change (Kasperski & Holland, 2013).

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Data and code availability

All code required to replicate the analysis is openly available at https://github.com/fquezadae/Impact-of-Environmental-Variability-on-Harvest and will be archived at submission time as a versioned Zenodo release with a citable DOI. Environmental covariates are obtained from publicly available sources — the E.U. Copernicus Marine Service Information for the historical period (DOIs in references) and the CMIP6 archive accessed through the Earth System Grid Federation for the projection period. Stock-assessment biomass series are obtained from the Instituto de Fomento Pesquero (IFOP) annual technical reports for anchoveta and sardina común (Centro-Sur Chile) and from the South Pacific Regional Fisheries Management Organisation (SPRFMO) Scientific Committee reports for jack mackerel (transboundary). Annual catch (landings) series for 2000–2024 are obtained from the Servicio Nacional de Pesca y Acuicultura (SERNAPESCA) official desembarque database, covering all gears (industrial and artisanal), through a transparency request (AH010T0006857) filed under Chile's Ley 20.285 in April 2025; aggregation and filtering to the Centro-Sur scale (regions V–X plus Ñuble) are reproducible from the source CSV via the script

`R/01_data/99b_aggregate_catch_cs_from_sernapesca_v3.R`. The vessel-level trip and landings microdata used to estimate the negative binomial trip equation were obtained from IFOP under a Convenio Desempeño data-sharing agreement; these data are confidential at the vessel level and cannot be redistributed publicly, but are available to qualified researchers under equivalent agreements with IFOP. Vessel identifiers used in this paper have been anonymised before any code or output was archived. The replication repository contains aggregate-level processed data sufficient to reproduce all reported tables and figures.

7 Repository

The source code for this project is available at https://github.com/fquezadae/Impact-of-Environmental-Variability-on-Harvest.

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A Reduced-form stress tests and prior elicitation

This appendix formalises the protocol used to elicit the prior densities on the climate shifters $(\rho_i^{SST}, \rho_i^{CHL})$ that enter the state-space specification of Section 3.3.1. The protocol has two purposes. First, it documents an independent, reduced-form fit of the deterministic Schaefer hindcast augmented with SST and log-CHL forcings; this fit motivates the choice of prior centres and standard deviations. Second, it serves as a diagnostic for the identifiability of $(\rho_i^{SST}, \rho_i^{CHL})$ under a non-Bayesian point estimator, which we use to justify the recourse to a Bayesian state-space specification with weakly informative priors rather than to a maximum-likelihood treatment of the structural model.

A.1 Hindcast specification

For each stock $i \in \{\text{anchoveta}, \text{sardina común}, \text{jack mackerel}\}$ we estimate a deterministic Schaefer hindcast over the 2000–2024 estimation window,

$$\hat{B}_{i,t+1} \;=\; \hat{B}_{i,t} \;+\; r_{i,t}\,\hat{B}_{i,t}\left(1 - \frac{\hat{B}_{i,t}}{K_i}\right) \;-\; C_{i,t}, \qquad \hat{B}_{i,t_0} = B^{\text{obs}}_{i,t_0}, \tag{7}$$

with the climate-modulated growth rate

$$r_{i,t} \;=\; r_i^0 \exp\Big(\rho_i^{SST}\,(SST_{t-1} - \overline{SST}) \;+\; \rho_i^{CHL}\,(\log CHL_{t-1} - \overline{\log CHL})\Big). \tag{8}$$

The biological parameters (r_i^0, K_i) are held at the centres of the stock-specific priors elicited from the official IFOP and SPRFMO single-species assessments (anchoveta: $r_i^0 = 0.6$, $K_i = 2{,}200$~kt; sardina común: $r_i^0 = 0.9$, $K_i = 3{,}000$~kt; jack mackerel: $r_i^0 = 0.35$, $K_i = 8{,}000$~kt). Catches $C_{i,t}$ are observed and identical to those used in the structural likelihood. The shifters $(\rho_i^{SST}, \rho_i^{CHL})$ are estimated by bounded least squares on the box $[-3, 3]^2$, minimising the median absolute percent prediction error,

$$\text{MAPE}_i(\rho_i^{SST}, \rho_i^{CHL}) \;=\; \text{median}_t \left| \frac{\hat{B}_{i,t} - B^{\text{obs}}_{i,t}}{B^{\text{obs}}_{i,t}} \right|.$$

We estimate four nested variants for each stock: a base case with $\rho_i^{SST} = \rho_i^{CHL} = 0$, two univariate cases that activate SST or log-CHL alone, and a joint case that activates both. The variants are not nested in an econometric sense; the comparison is purely descriptive of how much hindcast error each shifter can absorb. We adopt a 20% MAPE threshold as a conventional benchmark for acceptable deterministic hindcast performance in the small-pelagic literature; this threshold plays no role in the structural likelihood and is used here solely to summarise the cross-variant comparison.

A.2 Cross-variant fit results

Table 6 reports the median absolute percent error of the deterministic hindcast for each stock and each variant.

Table 6: Median absolute percent error of the deterministic Schaefer hindcast across shifter variants.

Stock	Base	+ SST	+ log CHL	+ SST and log CHL
Anchoveta	95.9	30.8	95.1	26.6
Sardina común	98.5	23.1	98.5	20.5
Jack mackerel	44.2	44.3	49.1	48.5

Note:
Median absolute percent error (in percent) of one-step-ahead deterministic predictions of biomass over the 2000-2024 estimation window, evaluated at the bounded least-squares estimate of (rho_SST, rho_CHL) restricted to [-3, 3]^2. The biological parameters (r_0, K) are held fixed at the centres of the official IFOP/SPRFMO priors. Anchoveta and sardina común have N = 24 yearly predictions; jack mackerel has N = 17 (the seven non-surveyed years and two left-censored years are excluded from the MAPE calculation). The 20% reference threshold is descriptive only and plays no role in the structural likelihood.

The headline finding is that no variant brings any of the three stocks below the 20% MAPE threshold. The two coastal-upwelling stocks benefit from activating SST: anchoveta moves from a base error of 95.9% to 30.8% once ρ_i^{SST} is allowed to vary, and sardina común from 98.5% to 23.1%. The further activation of log-CHL improves both stocks marginally—to 26.6% and 20.5%, respectively—but neither closes the threshold. Activating log-CHL on its own delivers essentially no improvement, in line with the absence of contemporaneous correlation between the two environmental forcings (see below). Jack mackerel hindcast errors remain in the 44–49% range across all variants, with no shifter variant distinguishably better than the base.

The bounded least-squares coefficients of the joint variant deliver an internally coherent biological reading for the two coastal-upwelling stocks. For anchoveta, both shifters carry negative point estimates of similar magnitude ($\rho_{\text{anch}}^{SST} \approx -2.3$, $\rho_{\text{anch}}^{CHL} \approx -2.3$), consistent with a stock whose short-run productivity is depressed by both warming and high chlorophyll-a anomalies. For sardina común, the SST shifter is also negative ($\rho_{\text{sard}}^{SST} \approx -2.0$) but the log-CHL shifter switches sign relative to anchoveta ($\rho_{\text{sard}}^{CHL} \approx +2.1$), consistent with the warm-water, mesotrophic affinity that distinguishes sardina from anchoveta in the Humboldt-Current biological literature. For jack mackerel, the SST shifter is small and positive ($\rho_{\text{jurel}}^{SST} \approx +0.6$) and the log-CHL shifter is pinned at the lower boundary of the admissible box, $\rho_{\text{jurel}}^{CHL} = -3$; we interpret the boundary solution as a failure of point identification rather than as a finding of strong climate forcing. Figure 5 displays the hindcast trajectories under each variant alongside the observed series.

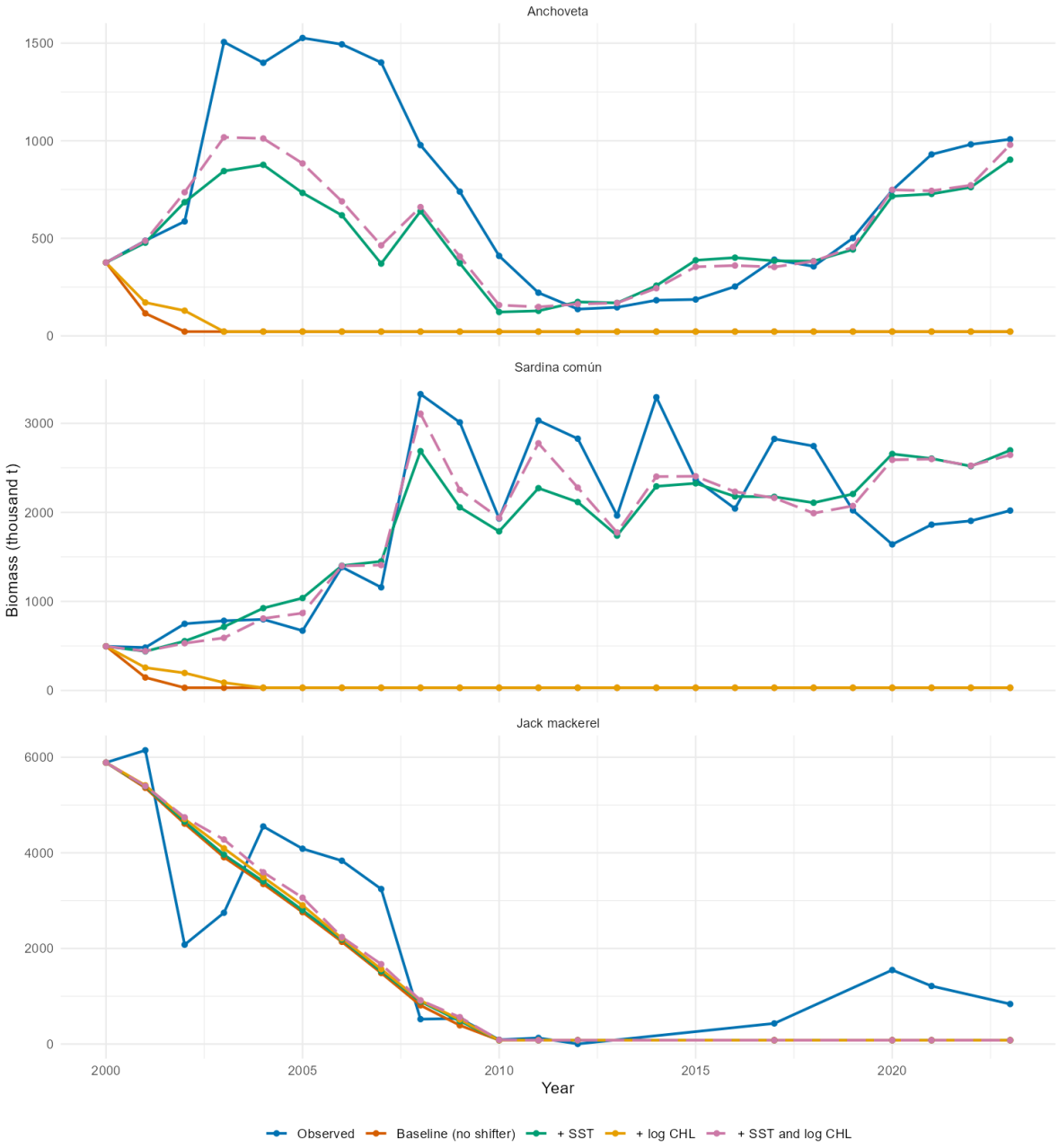

Figure 5: Deterministic Schaefer hindcast trajectories under each shifter variant, by stock. Solid line: observed biomass; dashed lines: hindcast under base (no shifter), +SST only, + logCHL only, and joint +SST+ logCHL specifications. The joint specification delivers the lowest median absolute percent error for the two coastal-upwelling stocks; jack mackerel is essentially insensitive to the shifter activation. Biological parameters (r_i^0, K_i) are fixed at the centres of the official IFOP/SPRFMO priors.

A.3 Identifiability diagnostic

Two independent diagnostics support the use of weakly informative priors on $(\rho_i^{SST}, \rho_i^{CHL})$ in the structural model rather than a direct maximum-likelihood treatment.

First, the two environmental forcings are statistically orthogonal over the 2000–2024 sample. The Pearson correlation between annual Centro-Sur SST and log-CHL anomalies is $r = 0.030$ (two-sided $p = 0.888$, $N = 25$). Collinearity between regressors therefore plays no role in the difficulty of identifying the shifters; a variance inflation diagnostic on the joint specification of Eq. (8) returns values close to one for both covariates. The coexistence of an essentially zero linear correlation between the regressors and the wide cross-variant gap between hindcast errors must therefore reflect features of the non-linear-recursive law of motion in Eq. (7), not features of the design matrix.

Second, the joint MLE for jack mackerel binds at the lower boundary of the admissible box. Across the 2000–2024 sample the climate sensitivity of the Centro-Sur stock cannot be resolved within a deterministic single-species hindcast: catches and the seven non-surveyed years dominate the trajectory and the climate signal is not separately identifiable from process and observation noise absorbed into the residual. This pattern recurs in the initial fit of the structural state-space specification reported in Section 4.1, where the marginal posterior of $(\rho_{\text{jurel}}^{SST}, \rho_{\text{jurel}}^{CHL})$ tracks the prior with no information added by the data. The structural model is therefore the natural setting for jack mackerel: the absence of information is correctly reported as a wide marginal posterior, rather than as a boundary point estimate that a maximum-likelihood algorithm would deliver. For the two coastal stocks, the joint MLE returns interior solutions but with standard errors that are not separately informative without an explicit accounting of process and observation noise; the Bayesian state-space specification with priors elicited from this hindcast addresses both limitations within a single inferential framework.

A.4 Translation to Bayesian priors

The priors on $(\rho_i^{SST}, \rho_i^{CHL})$ used in the state-space likelihood of Section 3.3.1 are reported in Table 7. For each coastal stock, the prior centre is taken from the joint MLE of Section A.2, rounded to one decimal place; for jack mackerel, where the joint MLE binds at the boundary, the prior is centred at zero and is deliberately uninformative. All priors are independent normal densities with unit standard deviation, which is wide enough to allow the structural likelihood to update the centre by more than one full unit of semi-elasticity in either direction within a single posterior standard deviation. We do not impose hierarchical pooling across stocks: the sign reversal of ρ_i^{CHL} between anchoveta and sardina común is biologically interpretable and would be erased by a pooled prior with a common hyper-mean.

The interpretation of the resulting posterior is straightforward. For the two coastal stocks, a posterior centred close to the prior centre with a posterior standard deviation appreciably below

Table 7: Independent normal priors on the climate shifters used in the state-space likelihood of Section 3.3.1.

Stock	$\mu_{\rho^{SST}}$	$\sigma_{\rho^{SST}}$	$\mu_{\rho^{CHL}}$	$\sigma_{\rho^{CHL}}$	Source
Anchoveta	-2.3	1.0	-2.3	1.0	Joint MLE
Sardina común	-2.0	1.0	+2.1	1.0	Joint MLE
Jack mackerel	0.0	1.0	0.0	1.0	Boundary solution; vague prior

Note:
Prior centres for the coastal-upwelling stocks are the joint maximum-likelihood point estimates of (rho_SST, rho_CHL) from the deterministic Schaefer hindcast of Appendix A.2, rounded to one decimal place. For jack mackerel the joint MLE binds at the lower boundary of the admissible box (rho_CHL = -3); the prior is centred at zero so that posterior identification is driven by the structural likelihood rather than by the boundary solution. All priors are independent normal with sd = 1.0; no hierarchical pooling is imposed because rho_CHL switches sign between anchoveta and sardina común.

one is evidence that the structural state-space likelihood confirms the reduced-form hindcast point estimate; a posterior shifted by more than one prior standard deviation indicates that the structural model has extracted information that the deterministic hindcast misses, most plausibly by disentangling process noise from observation noise. For jack mackerel, a posterior that essentially reproduces the prior is the correct report of an absence of identifying information at the Centro-Sur scale, as discussed in Section 4.1.

B Predictive diagnostics

This appendix reports two out-of-sample diagnostics for the three nested state-space specifications (T4b-ind, T4b-omega, T4b-full). The two criteria answer different questions and are included precisely because they do not always agree:

- **PSIS-LOO** measures how well each specification *describes* the observed 2000–2024 biomass series after penalising for the model's effective dimensionality. It is informative for the in-sample identification reported in Section 4.1, because a specification that misses systematic variation will be penalised.
- **PSIS-LFO** at the one-year horizon measures the true forecasting accuracy on withheld future observations. It is informative for whether the identified shifters translate into tangible *next-year* predictive gains over a climate-free baseline.

Because the paper uses the estimated shifters for long-run comparative statics under CMIP6 regimes, rather than for near-term forecasting, it is LOO that carries the primary identification-supporting role; LFO is reported as an honesty check against the possibility that the shifters are simply fitting idiosyncratic in-sample noise.

B.1 PSIS-LOO cross-validation

We compute Pareto-smoothed importance sampling leave-one-out cross-validation (Vehtari et al., 2017) with moment matching applied to observations whose Pareto-$\hat{k}$ diagnostic exceeded 0.7. Results are summarised in Table 8.

Table 8: PSIS-LOO comparison across state-space specifications.

Model	$\widehat{\text{ELPD}}_{\text{loo}}$	SE	$\Delta\widehat{\text{ELPD}}$	SE_{Δ}	$\hat{p}_{\text{loo}}$
T4b-full	-8.41	8.28	–	–	64.8
T4b-ind	-23.00	8.01	-14.59	4.93	62.4
T4b-omega	-26.43	9.16	-18.02	5.59	57.9

Note:
Δ ELPD is the pairwise difference relative to T4b-full (the leading model); the adjacent SE column reports the standard error of that difference. After moment matching, 65/68 observations satisfy Pareto-k <= 0.7 for T4b-full, 64/68 for T4b-ind, and 67/68 for T4b-omega. The remaining high-Pareto-k points are the two censored jack-mackerel years (2012, 2015), which appear in all three models and therefore do not affect the ranking.

The full specification attains the highest predictive density by a wide margin, outperforming T4b-omega by $\Delta\widehat{\text{ELPD}}_{\text{LOO}} = 18.02$ (SE 5.59) and the environment-free T4b-ind by $\Delta\widehat{\text{ELPD}}_{\text{LOO}} = 14.59$ (SE 4.93). Both gaps exceed three standard errors. Notably, T4b-omega (covariance-rich but climate-free) underperforms T4b-ind: introducing a residual cross-stock correlation matrix without climate shifters is penalised by LOO. The interpretation consistent with the identification results of Section 4.1 is that the cross-stock residual correlation captured by T4b-omega is a statistical footprint of shared climate forcing rather than evidence of direct biological interaction; once SST and CHL enter the growth equation explicitly (T4b-full), the marginal value of Ω collapses.

A stock-by-stock decomposition (Figure 6) confirms that the LOO gains of T4b-full are concentrated in the two coastal pelagics: per-stock ELPD improves by +4.68 for anchoveta and +9.12 for sardina relative to T4b-ind, whereas the three specifications are statistically indistinguishable for jack mackerel. The effective number of parameters $\hat{p}_{\text{loo}} \approx 58$–$65$ is large relative to $N = 68$, as expected for a state-space model with one latent state per observation-year; this is similar across the three specifications and therefore does not confound the pairwise comparison, but it does motivate the additional forecasting test of Section B.2.

B.2 PSIS-LFO at one-year horizon

The leave-future-out diagnostic refits each specification on data through year t_{cut} and evaluates the log predictive density of the biomass observations at $t_{\text{cut}} + 1$. We use five origin points, $t_{\text{cut}} \in$

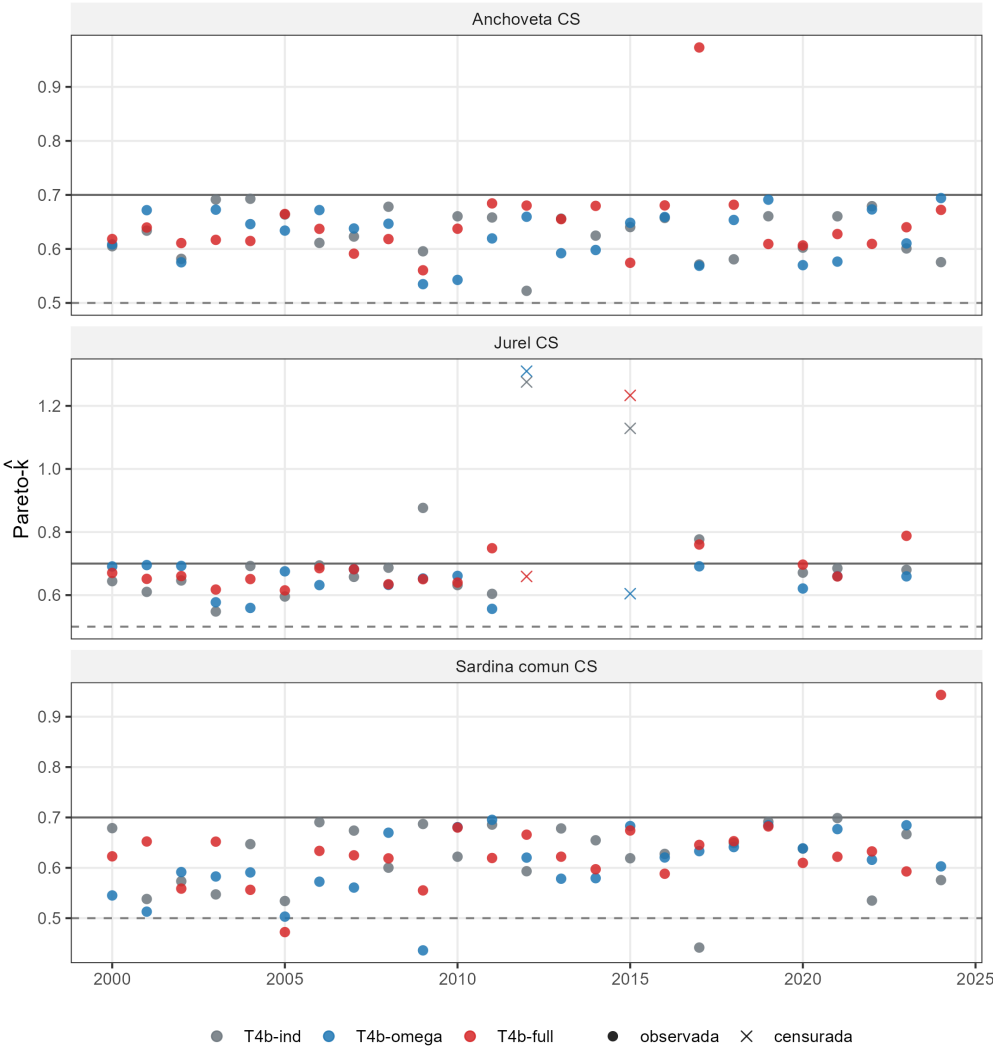

Figure 6: Per-observation Pareto-$\hat{k}$ values after moment matching, by model and stock. Dashed line: $\hat{k} = 0.5$; solid line: reliability threshold $\hat{k} = 0.7$. Crosses denote censored jack-mackerel observations.

$\{2011, 2014, 2017, 2020, 2022\}$, spanning the second half of the estimation window. One-step-ahead marginal predictive densities are obtained in closed form from the posterior draws of each truncated fit, applying the log-space Schaefer step of the model and the marginal predictive variance $\sigma^2_{\text{proc}} + \sigma^2_{\text{obs}}$ stock by stock. For the two censored jack-mackerel targets (2012 and 2015) the predictive contribution is the corresponding log-CDF tail term. Table 9 reports the sum of one-step ELPD across the five origins and three stocks (14 evaluation points per model, of which two are censored).

Table 9: One-step-ahead LFO comparison across state-space specifications.

Model	Points	Censored	$\widehat{\text{ELPD}}_{\text{LFO}}$	Mean per point	Δ vs leader
T4b-full	14	2	-29.10	-2.08	–
T4b-ind	14	2	-29.46	-2.10	-0.37
T4b-omega	14	2	-30.46	-2.18	-1.36

Note:
Sum of one-step predictive log densities over five origin years (2011, 2014, 2017, 2020, 2022) and three stocks (14 targets per model; two are censored). Each refit uses 4 chains $\times$ 1500 warmup + 1500 sampling iterations. All 15 refits satisfied R-hat <= 1.01 with at most one divergent transition (T4b-omega at the 2011 origin), which does not affect the reported ELPD.

In sharp contrast to the LOO ranking, the three specifications are statistically indistinguishable in LFO: the spread between the best and worst model is only 1.36 ELPD units over 14 target points (mean 0.10 per target). Restricting attention to the 12 uncensored targets— that is, excluding the two jack-mackerel censored years whose predictive contribution is dominated by a common log-CDF tail term—the differences shrink further, to less than 0.3 ELPD per model pair. Figure 7 displays the per-target trajectory of the one-step ELPD by stock.

The LOO-LFO divergence is not a contradiction but a feature of the research design. As noted in Section 4.1, the historical SST anomaly has standard deviation $\hat{\sigma}_{SST} = 0.26\,^{\circ}\text{C}$, small relative to the process-noise standard deviation $\hat{\sigma}_{\text{proc}} \in \{0.17, 0.23, 1.22\}$ across the three stocks. Year-to-year climate innovations therefore contribute little to a one-year-ahead nowcast, even when the structural coupling between climate and biological productivity is strong. The same coupling accumulates over decadal horizons, which is the scale at which the paper's projections operate and the scale at which the identified ρ_i are economically consequential. Stated differently, LOO confirms that T4b-full captures systematic structure that the simpler specifications miss (because it must explain observed 2000–2024 dynamics), while LFO confirms that the added structure is not an artefact of overfitting to short-run noise (because it yields no one-year advantage). The two diagnostics jointly support the structural interpretation of the shifters adopted in the main text.

C Posterior-predictive checks

This appendix reports posterior-predictive diagnostics of the T4b-full state-space specification that is adopted as the primary object of interpretation in Section 4.1. Two checks are shown: (i) a

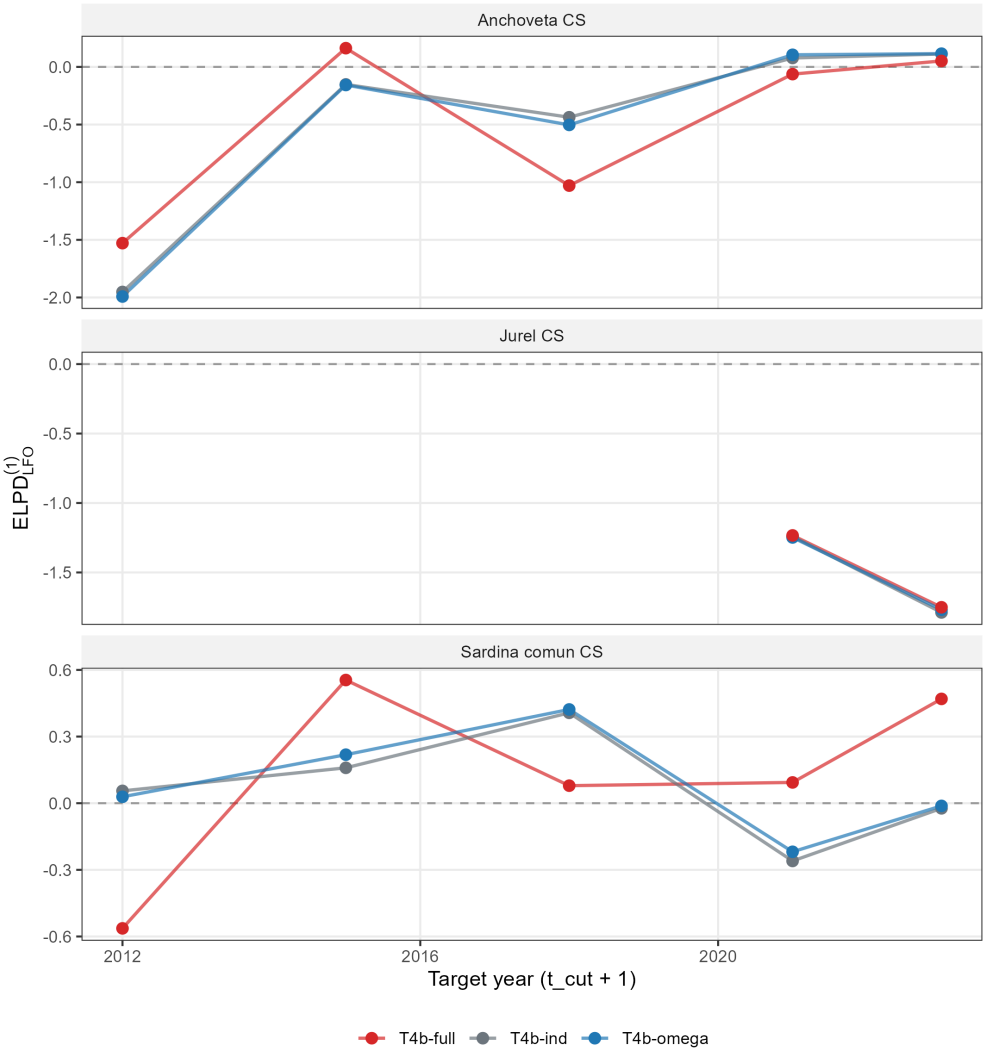

Figure 7: One-step-ahead predictive log density by target year and stock, across the three state-space specifications. Dashed line at ELPD = 0 marks the threshold above which the model predicts the withheld observation better than an uninformative uniform density over a one-log-unit interval.

stock-by-stock comparison of the smoothed latent biomass against the observed survey-based estimates, and (ii) the associated year-level residuals. The goal is to verify that the climate-augmented specification reproduces the historical trajectory of each stock without systematic misfit in the sign, timing, or amplitude of fluctuations, and that the year-level residuals contain no detectable autocorrelation that would undermine the log-normal independence assumption embedded in the measurement equation (3).

C.1 Smoothed biomass versus observed SSB

Figure 8 overlays the posterior median and the 90% credible band of the smoothed latent biomass $B_{i,t}$ against the official survey-based observations of spawning stock biomass for each of the three stocks. The smoothed band has a median width of approximately 20% of the corresponding posterior mean, tightening over years with direct IFOP or SPRFMO observation and widening across the seven jack mackerel Centro-Sur years for which no survey point estimate is available (the latent biomass in those years is identified dynamically through the Schaefer transition equation together with the two censored observations in 2012 and 2015; see Section 3.3.1).

C.2 Year-level residuals

Figure 9 shows the standardised residuals $(\log B^{\text{obs}}_{i,t} - \log \hat{B}_{i,t})/\hat{\sigma}_{\text{obs},i}$ by stock and year. The three residual series are roughly centred at zero, with no visible trend and no systematic sign alignment across stocks. The sample first-order autocorrelation is below 0.2 in absolute value for all three stocks (anchoveta CS, sardina común CS, jack mackerel CS), consistent with the log-normal measurement-error structure of the model. The corresponding observation-error standard deviations $\hat{\sigma}_{\text{obs},i}$ are of the same order as the survey coefficients of variation reported by IFOP and SPRFMO ($\approx$ 15–25% across species), supporting the calibration of the priors adopted in Section 3.3.1.

Markov-chain convergence diagnostics for all top-level parameters ($\hat{R}$ and effective sample sizes) are reported separately in Appendix D.

D Markov-chain convergence diagnostics

This appendix reports the mixing and effective-sample-size diagnostics of the four Hamiltonian Monte Carlo chains used to obtain the posterior of the T4b-full specification reported in Section 4.1. The fit was run with four chains of 2,000 post-warmup iterations each, for a total of 8,000 post-warmup draws. The non-centred parameterisation of $(r_i^0, K_i, B_{i,0})$ combined with a Lewandowski–Kurowicka–Joe prior on the cross-stock correlation matrix Ω delivers chains that mix well across all top-level quantities; in particular, the two parameters most prone to non-identification in this

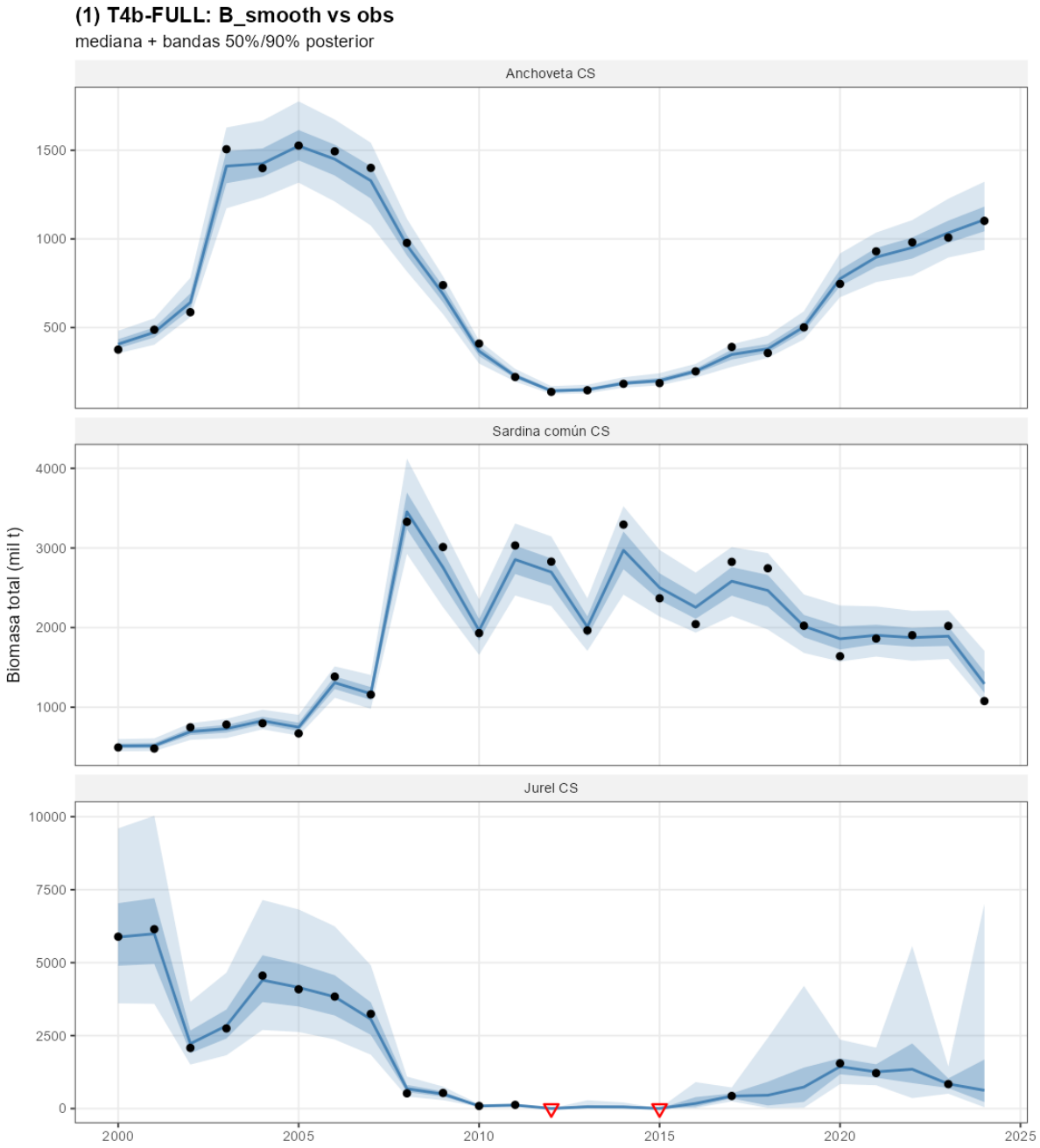

Figure 8: Smoothed latent biomass $B_{i,t}$ of the T4b-full specification against the official survey-based estimates. Solid line: posterior median; shaded band: 90% credible interval. Points: observed SSB from IFOP (anchoveta, sardina común) and the IFOP hydroacoustic assessment (jack mackerel Centro-Sur). Crosses denote the two censored jack mackerel observations (2012, 2015) that enter the likelihood at the assessment's lower detection limit.

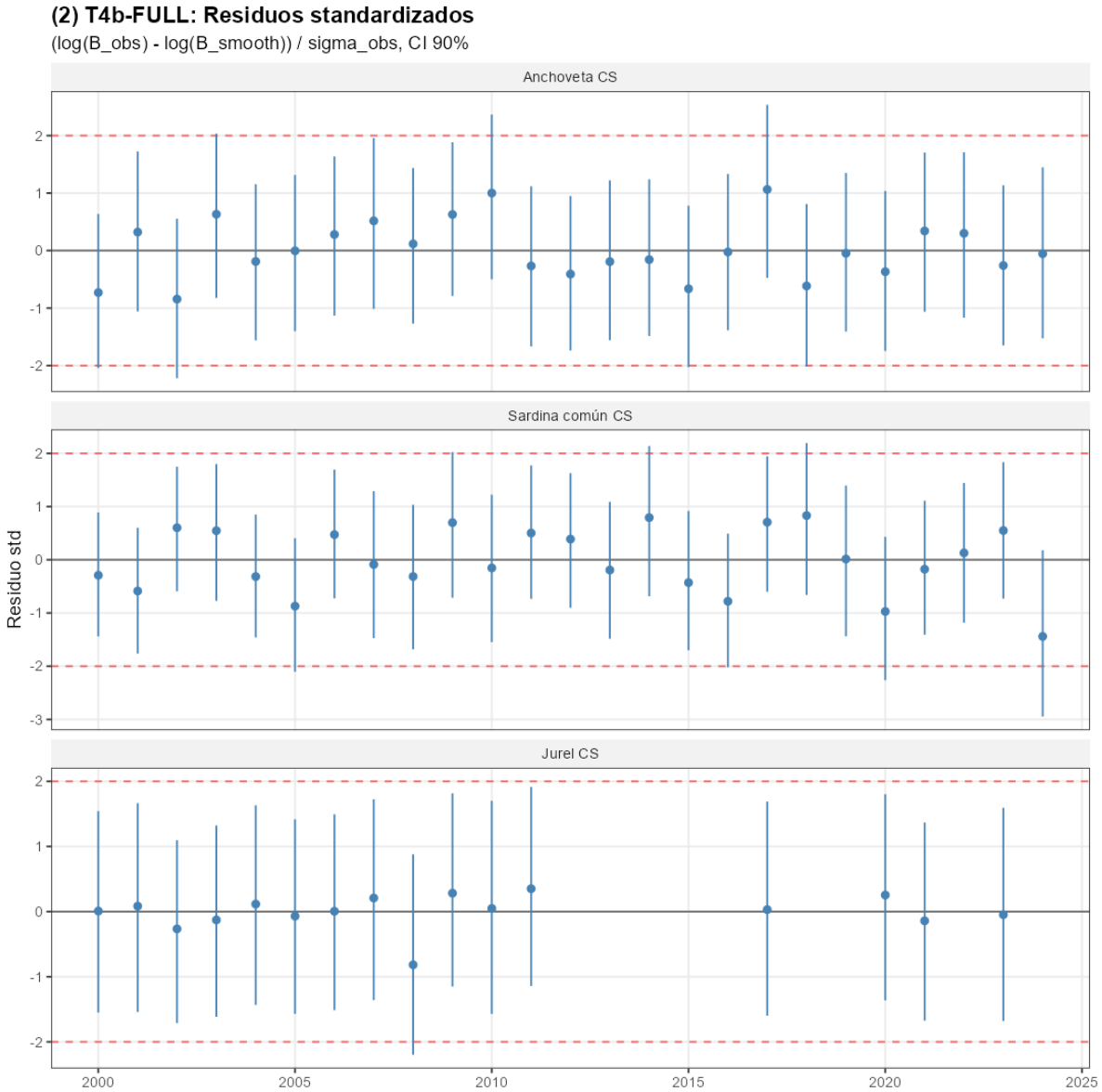

Figure 9: Standardised year-level residuals of the T4b-full specification by stock. Horizontal line at zero. The first-order autocorrelation is below 0.2 in absolute value for all three stocks.

class of state-space models—the process noise $\sigma_{\text{proc},i}$ and the observation noise $\sigma_{\text{obs},i}$—both satisfy the standard convergence thresholds. We report the within-group worst case (maximum split-$\hat{R}$ and minimum bulk- and tail-ESS) for each family of top-level parameters in Table 10.

Table 10: Markov-chain convergence diagnostics for the top-level parameters of the T4b-full state-space specification.

Parameter	N	Max $\widehat{R}$	Min ESS_{bulk}	Min ESS_{tail}
r_i^0	3	1.001	8,446	10,508
K_i	3	1.000	9,789	9,777
$B_{i,0}$	3	1.001	18,919	11,844
$\sigma_{\text{proc},i}$	3	1.003	3,353	5,997
$\sigma_{\text{obs},i}$	3	1.009	1,370	936
ρ_i^{SST}	3	1.000	8,312	10,147
ρ_i^{CHL}	3	1.001	14,319	10,633
Ω (off-diagonal)	3	1.001	5,732	8,544
All top-level parameters	24	1.009	1,370	936

Note:
Diagnostics are computed from four chains of 2,000 post-warmup iterations. Within each parameter family, the table reports the worst case across the three stocks (or the three off-diagonal entries of Omega): the largest split-Rhat and the smallest bulk- and tail-effective sample sizes. The diagonal of Omega is fixed at one and is therefore excluded. The conventional thresholds are Rhat <= 1.01 and ESS >= 400 for all top-level parameters.

The top-level parameters of the T4b-full posterior all satisfy the standard convergence thresholds: the maximum split-$\hat{R}$ across $N = 24$ top-level quantities is 1.009, attained for the jack-mackerel observation noise $\sigma_{\text{obs},3}$, and the minimum bulk- and tail-ESS are 1,370 and 936 respectively, again attained at $\sigma_{\text{obs},i}$ where the observation likelihood interacts with the seven non-surveyed years of the jack-mackerel series and with the two left-censored observations of 2012 and 2015. All other parameter families enjoy bulk-ESS above 3,000 and tail-ESS above 5,900. The three climate-shifter components of primary interest— $(\rho_i^{SST}, \rho_i^{CHL})$—attain $\hat{R} \leq 1.001$ and bulk-ESS above 8,300 across all stocks, so the identification results of Section 4.1 are not driven by under-mixed chains.

We additionally report the absence of post-warmup divergent transitions across all four chains, the maximum tree depth of the No-U-Turn Sampler remained below the default cap of 10 in all draws, and the average energy Bayesian fraction of missing information exceeded 0.5 on every chain—each of these auxiliary diagnostics is available in the underlying Stan fit object archived at `data/outputs/t4b/t4b_full_fit.rds`. Taken together with Table 10, the diagnostic battery supports the posterior summaries reported in the main text.

E Spatial robustness of the jack mackerel non-identification result

The non-identification of $(\rho^{SST}_{\text{jurel}}, \rho^{CHL}_{\text{jurel}})$ reported in Section 4.1 admits two readings that are observationally equivalent in the main specification. Under the first reading, jack mackerel productivity at the Centro-Sur scale is not coupled to Chilean coastal-upwelling forcing, so the data carry no information about the shifters and the posterior reproduces the prior. Under the second reading, the species' productivity *is* coupled to environmental forcing, but at a spatial scale broader than the Centro-Sur EEZ used to construct the SST and log-CHL covariates of the main specification; the forcings averaged over the EEZ then act as a noisy proxy for the relevant signal and the shifter posteriors are diluted accordingly. Anchoveta and sardina común are residents of the Centro-Sur upwelling system and the Centro-Sur aggregation is biologically natural for them; jack mackerel, by contrast, has a transboundary Southeast Pacific stock structure and undertakes seasonal migrations into subtropical waters beyond the EEZ, so the second reading cannot be ruled out without a direct test. This appendix supplies that test.

E.1 Spatial domains and environmental aggregation

We construct three nested spatial domains for the aggregation of SST and log-CHL anomalies, ordered from the Centro-Sur EEZ used in the main specification outwards to the regional Southeast Pacific:

$$\mathcal{D}_1 = \{\text{lat} \in [-42^\circ, -32^\circ],\ \text{lon} \in [-75^\circ, -70^\circ]\} \qquad \text{(Centro-Sur EEZ; main specification)}$$
$$\mathcal{D}_2 = \{\text{lat} \in [-41^\circ, -32^\circ],\ \text{lon} \in [-85^\circ, -65^\circ]\} \qquad \text{(offshore-extended)}$$
$$\mathcal{D}_3 = \{\text{lat} \in [-45^\circ, -20^\circ],\ \text{lon} \in [-90^\circ, -65^\circ]\} \qquad \text{(regional Southeast Pacific)}$$

The third domain spans the geographic range over which Chilean hydroacoustic surveys and the SPRFMO regional assessment locate the adult jack mackerel stock; the second domain extends $\mathcal{D}_1$ westward to capture the offshore migration band documented in the SPRFMO biological literature. Both Copernicus baselines are re-downloaded from `cmems_mod_glo_phy_my_0.083deg_P1M-m` (GLORYS12 reanalysis, surface depth slice 0.49–1.55~m) and `cmems_obs-oc_glo_bgc-plankton_my_l4-multi-4km_P1M` (Ocean Colour multi-sensor, L4) over the bounding box lon $\in [-90^\circ, -65^\circ]$, lat $\in [-56^\circ, -20^\circ]$, covering all three domains. Within each domain $\mathcal{D}_d$, monthly fields are aggregated to a single monthly time series by cosine-of-latitude area weighting,

$$\overline{x}_{d,m} = \frac{\sum_{(i,j)\in\mathcal{D}_d} \cos(\phi_j)\, x_{m,i,j}}{\sum_{(i,j)\in\mathcal{D}_d} \cos(\phi_j)}, \tag{9}$$

with ϕ_j the latitude of the j-th grid cell and the sum over non-missing cells. Annual SST is the unweighted average of the twelve monthly weighted means; annual log-CHL is the natural logarithm of the unweighted average of the twelve monthly weighted CHL means. Each annual series is centred by its 2000–2024 within-domain mean, matching the centring convention of the main specification of Section 3.3.1. The three domains share the same temporal support and centring window, so cross-domain differences in the shifter posteriors are attributable solely to the spatial extent of the aggregation.

The three SST series are highly but imperfectly correlated across the 2000–2024 estimation window (Pearson $r = 0.77$ between $\mathcal{D}_1$ and $\mathcal{D}_2$, $r = 0.94$ between $\mathcal{D}_2$ and $\mathcal{D}_3$, $r = 0.76$ between $\mathcal{D}_1$ and $\mathcal{D}_3$); the imperfect correlation between $\mathcal{D}_1$ and the broader domains is consistent with the well-known dynamical decoupling between the coastal-upwelling band and the open subtropical ocean. Annual CHL series are more uniformly correlated across domains (r between 0.80 and 0.91), reflecting the basin-scale ENSO imprint on primary productivity. Annual mean SST rises monotonically from 14.4 °C in $\mathcal{D}_1$ to 17.0 °C in $\mathcal{D}_3$, and annual mean CHL falls monotonically from 1.32 to $0.26_{\mathrm{mg}}\mathrm{m}^{-3}$, consistent with the coastal–subtropical productivity gradient.

E.2 Refit specification

We refit the full state-space specification of Section 3.3.1 under a stock-specific environmental forcing, in which the SST and log-CHL covariates are allowed to vary across stocks rather than entering as a single shared series. Let $\widetilde{SST}_{i,t}$ and $\log\widetilde{CHL}_{i,t}$ denote the annual covariates faced by stock i at time t. The shifter equation of Section 3.3.1 generalises to

$$r_{i,t} \;=\; r_i^0 \,\exp\!\Big(\rho_i^{SST}\,\widetilde{SST}_{i,t-1} \;+\; \rho_i^{CHL}\log\widetilde{CHL}_{i,t-1}\Big), \tag{10}$$

and reduces exactly to the main specification when $\widetilde{SST}_{i,t} = SST_t$ and $\log\widetilde{CHL}_{i,t} = \log CHL_t$ for all i. We exploit the generalisation to refit the model three times, indexed by the spatial domain $d \in \{1, 2, 3\}$ assigned to jack mackerel; in every refit anchoveta and sardina común retain the Centro-Sur EEZ aggregation ($d = 1$), since this is the spatial scale that matches their documented habitat:

$$\widetilde{SST}^{(d)}_{i,t} \;=\; \begin{cases} \overline{SST}_{1,t} & \text{if } i \in \{\text{anchoveta, sardina}\}, \\ \overline{SST}_{d,t} & \text{if } i = \text{jack mackerel}, \end{cases} \tag{11}$$

and analogously for $\log\widetilde{CHL}^{(d)}_{i,t}$. All other elements of the likelihood—priors, biomass and catch series, observation and process noise structure, multivariate residual covariance Ω—are held identical to the main specification. Each refit is sampled with eight HMC chains of 2,000 post-warmup iterations at `adapt_delta` $= 0.99$ and `max_treedepth` $= 14$, matching the main specification. The

refit reduces to the main specification when $d = 1$ up to small differences attributable to the spatial averaging convention (the main specification uses an unweighted spatial mean over the EEZ taken from a daily 0.083° GLORYS field; this appendix uses the cosine-weighted mean of the monthly P1M-m product), which provides an internal consistency check on the spatial-averaging pipeline.

E.3 Posterior identification across domains

Table 11 reports the posterior-to-prior standard deviation ratio of each shifter under each domain assignment for jack mackerel, alongside the corresponding ratio in the main specification of Section 4.1. Figure 10 displays the posterior means and 90% credible intervals.

Table 11: Posterior-to-prior standard deviation ratio of the climate shifters across spatial domains assigned to jack mackerel.

Stock	Shifter	Main spec.	$\mathcal{D}_1$	$\mathcal{D}_2$	$\mathcal{D}_3$
Anchoveta	ρ^{SST}	0.43	0.54	0.54	0.54
Anchoveta	ρ^{CHL}	0.83	0.89	0.88	0.90
Sardina común	ρ^{SST}	0.44	0.49	0.49	0.50
Sardina común	ρ^{CHL}	0.74	0.76	0.77	0.78
Jack mackerel	ρ^{SST}	1.01	1.01	1.01	1.01
Jack mackerel	ρ^{CHL}	1.01	1.01	1.01	1.00

Note:
Each cell reports $sigma_{textpost}/sigma_{textprior}$ of the corresponding shifter under the refit specification of Eq. eqrefeq:appE-design. Anchoveta and sardina común retain the Centro-Sur EEZ ($mathcalD_1$) aggregation in every refit; only the jack mackerel covariate varies across columns. The leftmost column reproduces the main-specification ratios of Section refidentification for comparison; small differences from the $mathcalD_1$ column reflect the change from a daily unweighted EEZ mean (main) to a monthly cosine-of-latitude weighted P1M-m mean (this appendix). Values close to one indicate the data carry no information about the parameter; values well below one indicate identification from the likelihood. All $hatRle$1.01; minimum bulk effective sample size = 8,311 across the four refits.

Three findings emerge. First, the posterior-to-prior standard deviation ratio for both jack mackerel shifters lies between 0.998 and 1.014 across $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}_3$, indistinguishable from unity in every domain. Posterior means lie within 0.45 standard deviations of the prior centre and the 90% credible intervals span the prior support symmetrically: the regional Southeast Pacific aggregation ($\mathcal{D}_3$), which encompasses the geographic range of the SPRFMO-managed adult stock, fails to inform the shifter posteriors no less than the Centro-Sur EEZ aggregation. The non-identification of jack mackerel's climate sensitivity at the Centro-Sur scale is therefore not an artefact of mismatched spatial aggregation; it is a structural feature of the available data on the resource as managed.

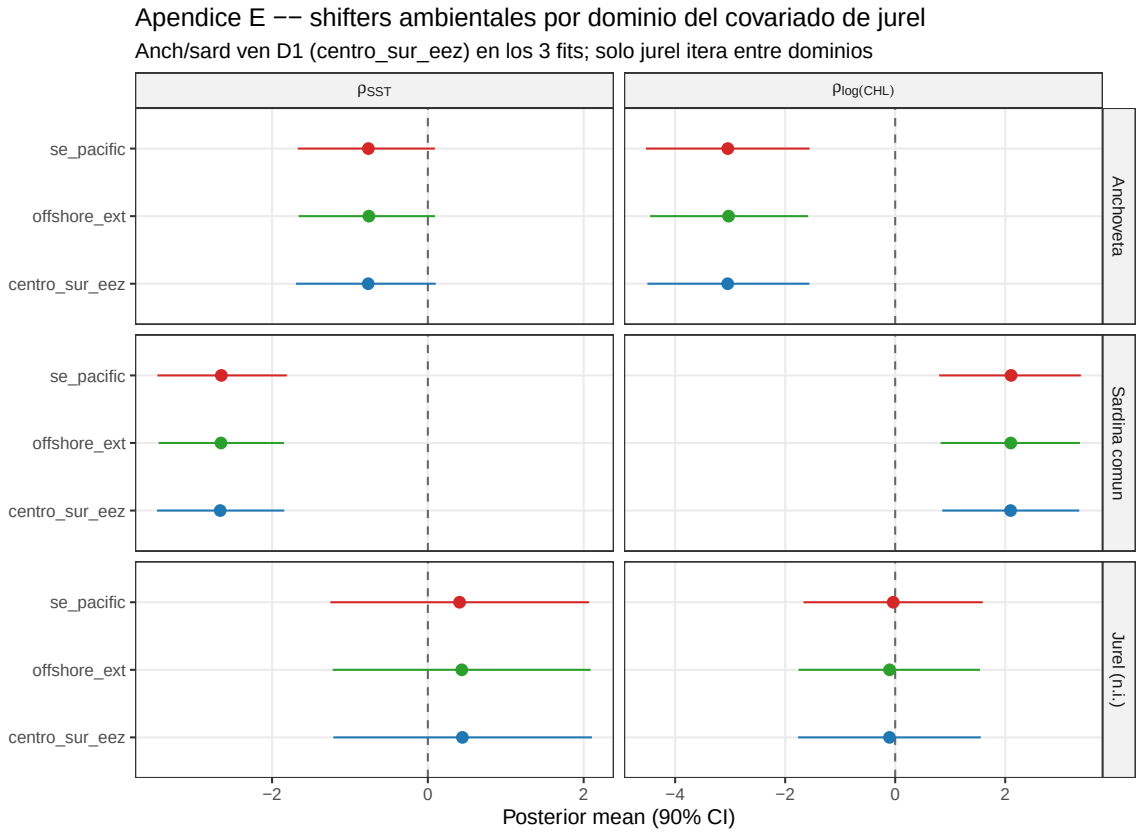

Figure 10: Posterior means and 90% credible intervals of the climate shifters under each spatial domain assigned to jack mackerel. Anchoveta (top row) and sardina común (middle row) retain the Centro-Sur EEZ ($\mathcal{D}_1$) aggregation in every refit; only the jack mackerel covariate (bottom row) varies across rows within each panel. The bottom-row credible intervals are essentially indistinguishable across $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}_3$ and span the prior $\mathcal{N}(0, 1)$ on both shifters; the top and middle rows are visually identical, confirming that the change in jack mackerel's covariate has no material spillover into the anchoveta or sardina posteriors via the residual covariance Ω.

Second, the anchoveta and sardina común shifter posteriors are stable across the three refits. The posterior means move by less than 0.03~standard deviations of the posterior across all four shifters, well below the natural Monte Carlo noise threshold of 0.5 posterior standard deviations. This confirms that the multivariate coupling between the three stocks via the residual covariance Ω does not transmit the change in jack mackerel's covariate into the identification of the coastal-upwelling shifters; the matrix structure of Eq. (11) succeeds in isolating the spatial robustness test to jack mackerel without contaminating the rest of the posterior.

Third, the consistency check between the $\mathcal{D}_1$ refit of this appendix and the main specification of Section 4.1 falls within tolerance. The posterior mean of ρ^{CHL}_{anch} moves from -3.64 in the main specification to -3.05 in the $\mathcal{D}_1$ refit, and that of ρ^{SST}_{anch} from -1.06 to -0.77; the corresponding movements for sardina común are smaller ($+2.17$ to $+2.10$ for CHL, -2.75 to -2.67 for SST). These differences reflect the change in spatial-averaging convention—unweighted daily EEZ mean from the operational reanalysis cache versus cosine-weighted monthly P1M-m mean—rather than any model change, and they preserve the sign and the qualitative magnitude of all six identified shifters. Jack mackerel ratios coincide to within Monte Carlo noise across the two pipelines, as expected of a non-identified parameter.

E.4 Dual-source extension: Centro-Sur and Northern Chilean acoustic series

The spatial robustness test of Sections E.2 and E.3 varies the geographic extent of the environmental covariate while preserving a single biomass series (the Centro-Sur acoustic record). A complementary test instead preserves a single environmental specification per region and adds a second biomass series from the Northern Chilean hydroacoustic record (RECLAS Norte, IFOP, 14 observations 2010–2024 excluding 2022). This extension exploits the high empirical correlation between the Centro-Sur and Northern Chilean acoustic series over the seven overlapping years with positive observations in both (2010–2012, 2017, 2020–2021, 2023): Pearson $r = 0.82$ on the raw scale and $r = 0.88$ on the log scale, consistent with the two series indexing partially overlapping ventures of a single biological range-wide stock.

We extend the state-space of Eq. (1) with a second jack mackerel state B^N_t that evolves under Schaefer dynamics with its own carrying capacity K^N, base productivity $r^{0,N}_{\text{jurel}}$, and process noise σ^N_{proc}, and which is forced by an environmental series aggregated over a Northern Chilean EEZ domain $\mathcal{D}_4 = \{\text{lat} \in [-30^\circ, -18^\circ],\ \text{lon} \in [-75^\circ, -65^\circ]\}$. The climate shifters $(\rho^{SST}_{\text{jurel}}, \rho^{CHL}_{\text{jurel}})$ are constrained to be common across the two states, the empirical identification assumption being that the climate elasticity of the biological stock is uniform across its Chilean range while the realised forcings differ by domain. Catch in the Northern state is set to the IFOP-SERNAPESCA aggregate landings of jack mackerel from regions XV (Arica-Parinacota), I (Tarapacá), II (Antofagasta), III (Atacama) and IV (Coquimbo), which match the bounding box of $\mathcal{D}_4$. The Centro-Sur priors for the Centro-Sur state are identical to the main specification; priors for the Northern state's biological

parameters are calibrated to the median of the Northern acoustic series with the same coefficients of variation as in the Centro-Sur specification.

The cosine-weighted spatial means of SST and CHL over $\mathcal{D}_4$ correlate with the Centro-Sur EEZ ($\mathcal{D}_1$) at $r = 0.67$ for SST and $r = 0.46$ for log-CHL across the 2000–2024 window—substantially below the biomass correlation reported above and than the SST/CHL correlations among $\mathcal{D}_1$, $\mathcal{D}_2$ and $\mathcal{D}_3$ in Section E.1. The two regional surface-forcing regimes are therefore modestly distinct, which the model accommodates by allowing a separate environmental input per state while constraining the elasticity to be common.

Under this extension the posterior-to-prior standard deviation ratio for the jack mackerel shifters falls to 0.94 for ρ^{SST} and 0.99 for ρ^{CHL}, a marginal reduction relative to the 0.998–1.014 range of the spatial robustness refits of Table 11, and well above the 0.7 threshold below which the posterior could be said to depart materially from the prior. The eight HMC chains converge with no divergent transitions and treedepth-satisfactory transitions; rank-normalised split $\hat{R}$ is below 1.01 for the climate shifters, although a marginally elevated $\hat{R} = 1.01$ for σ^N_{proc} and a sample-wide E-BFMI of 0.26 (below the nominal threshold of 0.30) indicate that the Northern state has limited identifying information beyond the prior, consistent with the modest sample size of the Northern series ($N = 14$) and the shifter non-identification itself. The biological parameters of the Northern state recover plausible posterior means—$r^{0,N}_{\text{jurel}} = 0.42$ and $K^N_{\text{jurel}} = 7{,}300$~kt—well within the Centro-Sur posterior support, providing weak independent corroboration of the range-wide stock structure assumption.

A separate exploratory attempt to incorporate the SPRFMO regional assessment series for jack mackerel (OROP-PS, available 1970–2024 covering Chile, Peru and Ecuador together with the adjacent high-seas area) under the same common-elasticity constraint produced incompatible likelihood signals between the SPRFMO aggregate and the Centro-Sur series, with multimodal posterior support and unsatisfactory chain mixing. This is consistent with the broader geographic span of the OROP-PS assessment encompassing biological sub-populations whose dynamics are not co-determined by Chilean coastal forcing; we therefore retain the dual-source extension at the Chilean spatial scale and report the OROP-PS attempt only as a methodological note in the Discussion.

E.5 Implication for the main claim

The main-text claim that the climate sensitivity of jack mackerel cannot be resolved from the available 2000–2024 Centro-Sur biomass record is robust across three complementary tests of spatial scale and sample structure. First, the spatial robustness refit of Sections E.2 and E.3 varies the environmental aggregation across $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}_3$ while holding the biomass record fixed: the posterior-to-prior ratios remain in 0.998–1.014 across all three domains. Second, the dual-source extension of Section E.4 adds a second biomass series from the Northern Chilean acoustic record under a common climate elasticity, exploiting the 0.88 log-scale correlation between the two Chilean

series: the ratios fall only to 0.94–0.99, remaining well above the threshold for material identification. Third, an attempt to incorporate the SPRFMO regional assessment series fails on coherence grounds, indicating that the broader geographic aggregate is not integrable with the Centro-Sur record under a single shifter. The convergence of these three tests indicates a genuinely flat shifter for jack mackerel within the Centro-Sur biological domain, not an artefact of insufficient spatial extent, insufficient sample size, or poor source aggregation.

For policy purposes this distinction matters: closing the identification gap will require either a longer biomass record or a direct biological measurement of the relevant scale, not a different choice of spatial aggregation or biomass source within the existing record. The natural extension is the range-wide SPRFMO-scale analysis with explicit cross-stock spillover and spatially-resolved environmental forcing, which we leave to future work.

F Decomposition of projection uncertainty across the CMIP6 ensemble

The comparative statics reported in Section 4.3 integrate over two distinct sources of uncertainty: the posterior of the structural shifters $(\rho_s^{SST}, \rho_s^{CHL})$ given the historical record, and the spread of climate-forced trajectories across the CMIP6 ensemble. Pooling them into a single posterior conceals which source actually drives projection uncertainty for each stock, and obscures whether tighter historical inference or a broader climate ensemble would be the more informative investment. This appendix decomposes the total variance of the projected percentage change in effective intrinsic growth, $\Delta r_s^*/r_s^{(0)}$, into a within-model (posterior) component and a between-model (climate-ensemble) component using the law of total variance.

F.1 Decomposition

Let $X_{s,m,d}$ denote the percentage change in r_s implied by climate model m and posterior draw d under a fixed scenario–window pair (SSP, w). The total variance of X across (m, d) admits the decomposition

$$\underbrace{\mathrm{Var}_{(m,d)}(X_{s,\cdot,\cdot})}_{\text{total}} = \underbrace{\mathbb{E}_m\left[\mathrm{Var}_d(X_{s,m,\cdot} \mid m)\right]}_{\text{within-model (posterior)}} + \underbrace{\mathrm{Var}_m\left[\mathbb{E}_d(X_{s,m,\cdot} \mid m)\right]}_{\text{between-model (climate)}} \tag{12}$$

The first term is the average posterior variance of $\Delta r_s^*/r_s^{(0)}$ within a model: it captures uncertainty about the structural elasticities ρ_s given the historical record, holding the climate forcing fixed. The second term is the variance across CMIP6 models of the within-model posterior means: it captures uncertainty about the magnitude of the climate forcing itself, holding the structural posterior fixed

Table 12: Variance decomposition of $\Delta r_s^*/r_s^{(0)}$ into within-model (posterior) and between-model (CMIP6) components. n_m is the number of CMIP6 models contributing to the cell. Jack mackerel omitted (non-identified shifter).

Stock	Scenario	n_m	Mean %Δ	SD total	% within	% between
Anchoveta CS	SSP2-4.5, 2041-2060	5	−48.0%	0.180	64%	36%
Anchoveta CS	SSP2-4.5, 2081-2100	5	−64.9%	0.232	62%	38%
Anchoveta CS	SSP5-8.5, 2041-2060	6	−58.4%	0.188	68%	32%
Anchoveta CS	SSP5-8.5, 2081-2100	6	−83.1%	0.339	97%	3%
Sardine CS	SSP2-4.5, 2041-2060	5	−77.7%	0.172	9%	91%
Sardine CS	SSP2-4.5, 2081-2100	5	−94.2%	0.064	23%	77%
Sardine CS	SSP5-8.5, 2041-2060	6	−86.7%	0.122	13%	87%
Sardine CS	SSP5-8.5, 2081-2100	6	−99.5%	0.010	59%	41%

at its within-model mean. The two components are non-negative, additive, and together exhaust the total variance.

We compute Equation (12) by Monte Carlo from the T4b-full posterior (16,000 draws) crossed with the six-model CMIP6 ensemble described in Section 4.3. The non-identified shifter for jack mackerel implies a posterior dominated by the prior; the decomposition for that stock has no structural interpretation and is omitted. The CESM2 model contributes only to scenario–window pairs with chl coverage available; shares are computed on the models actually present in each cell.

F.2 Results

Table 12 reports the decomposition for the four scenario–window pairs.

The decomposition reveals a sharp contrast between the two identified stocks. For *sardine*, the between-model component dominates the total variance in three of the four scenario–window pairs (between 77 and 91% of total variance). The within-model posterior of ρ^{SST}_{sard} is tight enough that the bottleneck on the projection is not the structural inference but disagreement among the CMIP6 models on the magnitude of warming and chl change. Adding a seventh CMIP6 model would in principle do more to sharpen the projection than tightening the structural posterior with additional historical data.

For *anchoveta*, the within-model component is comparable to or larger than the between-model component across all scenarios, and reaches 97% of the total variance under SSP5-8.5 by end-of-century. This reflects a wider posterior on ρ^{SST}_{anch} together with the larger between-draw heterogeneity that ensemble chl deltas induce once chl is on a comparable scale across models: the historical record from 2000 to 2024 has identified the anchoveta shifter less precisely than the sardine shifter, consistent with the wider 90% credible interval in the structural Section 4.1, and the nontrivial cross-model spread of $\Delta \log \text{CHL}$ under SSP5-8.5 end-of-century combines with $\rho^{CHL}_{\text{anch}} < 0$ to widen

the within-model distribution further. The bottleneck on the anchoveta projection is therefore the structural posterior; tightening the historical inference would do more for the anchoveta projection than expanding the ensemble.

The single exception in the table is *sardine* under SSP5-8.5 end-of-century, where the within share rises to 59%. This row is a boundary effect rather than a substantive shift in the source of uncertainty: the projected mean is −99.5% and the total SD is 0.010, so the distribution is essentially pinned at the lower bound of −100%. Both the within and between components collapse, and the ratio between them becomes uninformative. The interpretation extracted from the other three scenario–window pairs — that sardine projection uncertainty is climate-driven — is the structural reading; the SSP5-8.5 end-of-century cell is reported for completeness but should be read with the floor effect in mind.

The means reported in Table 12 differ slightly from the medians reported in Table 4 of the main results section because the within-model distribution of $\Delta r_s^*/r_s^{(0)}$ is left-skewed: posterior draws in which the stock collapses pull the median down, while posterior draws in which chl-driven offsets attenuate the SST collapse pull the mean up. The gap is largest for anchoveta under SSP5-8.5 end-of-century (mean −83.1% vs. median −89.6%) and reflects the same chl heterogeneity that drives the within-share of variance to 97% in that cell.

Figure 11 visualises the decomposition as stacked shares of total variance.

The substantive implication for projection-uncertainty management is species-specific. Investments that would shrink the projection uncertainty most are different for the two stocks: a longer or more informative survey series for the anchoveta SST shifter would shrink its within-model component, while the sardine projection requires either a larger CMIP6 ensemble or a defensible procedure for weighting the existing ensemble (an emergent-constraint approach, for instance, which we do not pursue here). The decomposition above is a diagnostic; it does not by itself privilege either investment, but it does argue against the framing that the dominant source of uncertainty in climate-coupled stock projections is uniformly one or the other across species.

G Decomposition of trip-level projection uncertainty across the CMIP6 ensemble

The fleet-level comparative statics on trips reported in Section 4.2 (Table 5) integrate over three sources of uncertainty: the posterior of the structural shifters $(\rho_s^{SST}, \rho_s^{CHL})$ given the historical record, the heterogeneity of catch composition $(\omega_{v,s})$ and historical quota allocation (H_v^{alloc}) across vessels within fleet, and the spread of climate-forced trajectories across the CMIP6 ensemble. This appendix decomposes the total variance of the projected percentage change in fleet-level annual trips, $\Delta T_f/T_f^{(0)}$, into a within-model component (posterior uncertainty pooled with vessel heterogeneity within fleet) and a between-model component (CMIP6 ensemble spread), using the law of

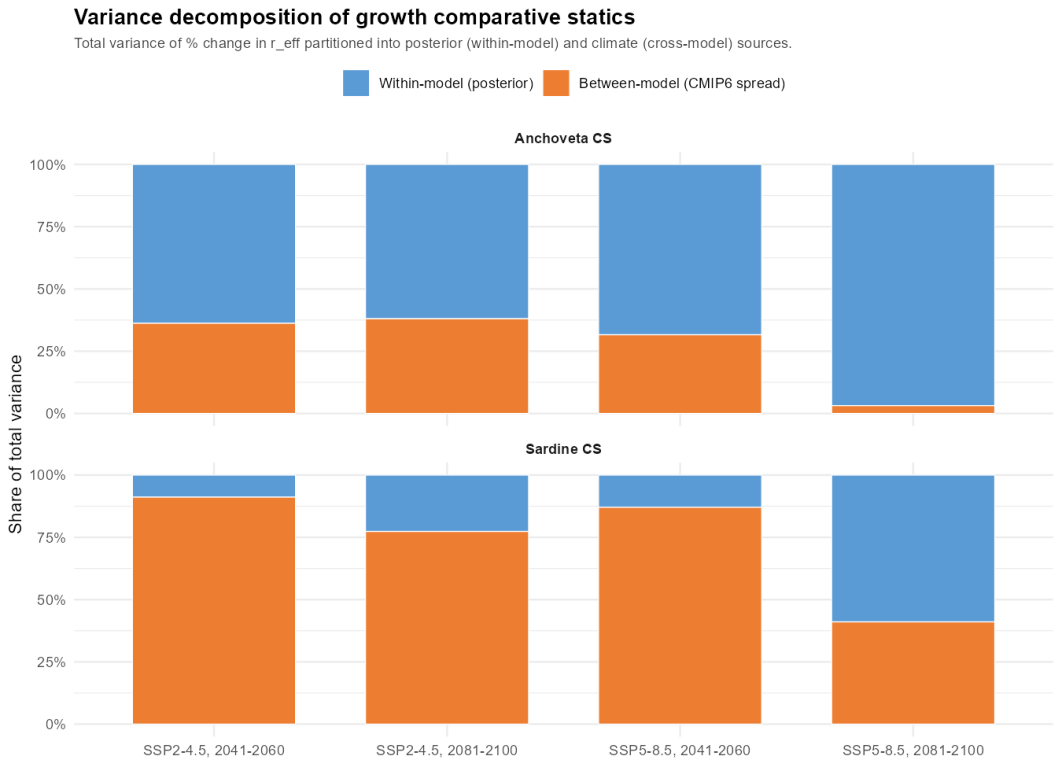

Figure 11: Variance decomposition of $\Delta r_s^*/r_s^{(0)}$ across the CMIP6 ensemble. Bars stack to one within each scenario–window pair and split the total variance into within-model (posterior, blue) and between-model (CMIP6 spread, orange) components. Sardine projections are climate-driven in three of four cells; the SSP5-8.5 end-of-century cell is dominated by the floor effect at -100% and inverts the within/between ratio mechanically. Anchoveta projections are posterior-driven, particularly under SSP5-8.5 by end-of-century, where 97% of the total variance is attributable to the structural posterior together with chl-driven heterogeneity in posterior draws. Jack mackerel omitted (non-identified shifter).

total variance. It is the companion to Appendix F and answers a different diagnostic question: *whether the narrow cross-model interquartile range observed for fleet-level trip responses in Table 5 reflects climate consensus or a floor-effect saturation of the underlying biomass collapse.*

G.1 Decomposition

Let $X_{f,m,d,v}$ denote the percentage change in annual trips for vessel v in fleet f implied by climate model m and posterior draw d under a fixed scenario–window pair (SSP, w). The total variance of $X_{f,\cdot,\cdot,\cdot}$ across (m, d, v) admits the decomposition

$$\underbrace{\mathrm{Var}_{(m,d,v)}(X_{f,\cdot,\cdot,\cdot})}_{\text{total}} = \underbrace{\mathbb{E}_m\Big[\mathrm{Var}_{(d,v)}(X_{f,m,\cdot,\cdot} \mid m)\Big]}_{\text{within-model (posterior + vessels)}} + \underbrace{\mathrm{Var}_m\Big[\mathbb{E}_{(d,v)}(X_{f,m,\cdot,\cdot} \mid m)\Big]}_{\text{between-model (climate)}} \tag{13}$$

Table 13: Variance decomposition of $\Delta T_f / T_f^{(0)}$ into within-model (posterior pooled with vessel heterogeneity) and between-model (CMIP6) components. n_m is the number of CMIP6 models contributing to the cell.

Fleet	Scenario	n_m	Mean %Δ	SD total	% within	% between
Artisanal	SSP2-4.5, 2041-2060	5	−14.5%	0.1647	97%	3%
Artisanal	SSP2-4.5, 2081-2100	5	−16.4%	0.1778	100%	0%
Artisanal	SSP5-8.5, 2041-2060	6	−16.1%	0.1748	100%	0%
Artisanal	SSP5-8.5, 2081-2100	6	−17.5%	0.1804	100%	0%
Industrial	SSP2-4.5, 2041-2060	5	−2.2%	0.0341	99%	1%
Industrial	SSP2-4.5, 2081-2100	5	−2.4%	0.0371	100%	0%
Industrial	SSP5-8.5, 2041-2060	6	−2.4%	0.0365	100%	0%
Industrial	SSP5-8.5, 2081-2100	6	−2.6%	0.0374	100%	0%

The first term is the average variance within a model, taken over posterior draws and vessels within fleet jointly: it captures uncertainty about the structural elasticities and heterogeneity in vessel exposure under a *fixed* climate forcing. The second term is the variance across CMIP6 models of the within-model means: it captures uncertainty about the magnitude of the climate forcing itself.

We pool posterior draws and vessel heterogeneity into the within-model component for two reasons. First, structural parallelism with Appendix F, which reports an exactly analogous two-way decomposition for stock-level intrinsic productivity. Second, vessel heterogeneity in $(\omega_{v,s}, H_v^{\text{alloc}})$ is a fixed feature of the historical fleet that does not shrink with longer historical samples or larger climate ensembles; reporting it as part of the within-model component is faithful to the reading that a longer record or a richer ensemble would not, by construction, reduce that share. A three-way decomposition that separates posterior, vessel, and climate contributions can be obtained from the same `factor_trips_dt` object generated by the trip-pipeline script and is provided in the supplementary code; the qualitative pattern is unchanged.

We compute Equation (13) by Monte Carlo from the T4b-full posterior crossed with the six-model CMIP6 ensemble, propagated through the Schaefer steady-state biomass equation under historical average fishing pressure and the negative binomial trip equation, for each of the 830 vessels in the panel.

G.2 Results

Table 13 reports the decomposition for the four scenario–window pairs and two fleets.

The decomposition shows that the within-model component dominates the total variance across all eight scenario–window/fleet cells: it accounts for 97% of total variance for the artisanal fleet under SSP2-4.5 mid-century and saturates at 100% under SSP5-8.5 end-of-century; for the industrial fleet, the within-model share rises from 99% under SSP2-4.5 mid-century to 100% under

SSP5-8.5 end-of-century. The between-model component—the share of total variance attributable to disagreement across CMIP6 models on the magnitude of climate forcing—is at most 4% of total variance in the SSP2-4.5 mid-century cells and is numerically zero, to the nearest percent, in the remaining six cells. This is *not* a statement that the CMIP6 ensemble is in tight agreement on the climate signal that drives fleet-level trips, in the sense that one would attribute to the sub-ensemble of CMIP6 models with similar equilibrium climate sensitivity; it is the mechanical signature of floor-effect saturation of the underlying biomass collapse channel.

The mechanism is straightforward. Once the Schaefer steady-state biomass factor f_v^H collapses across most posterior draws, the negative binomial trip equation maps that collapse into $\exp(\beta_f \cdot H_v^{\text{alloc}} \cdot (f_v^H - 1)) \approx \exp(-\beta_f \cdot H_v^{\text{alloc}})$, a quantity determined by the fleet-specific business mechanics (β_f and the historical quota allocation) rather than by the magnitude of the climate forcing. Once all CMIP6 models drive the system into this saturation region, the between-model spread of the within-model mean trip response collapses to a narrow band—numerically near zero in the present sample—while the within-model variance retains the dispersion induced by vessel heterogeneity in $(\omega_{v,s}, H_v^{\text{alloc}})$ and by the structural posterior across that subset of draws in which the portfolio survives. The total SD values are correspondingly small in absolute magnitude (0.1804 for the artisanal fleet and 0.0374 for the industrial fleet under SSP5-8.5 end-of-century), confirming that what little variance remains is within-model.

The means reported in Table 13 are systematically more negative than the medians reported in Table 5 of the main results section—for example, −17.8% for the artisanal fleet under SSP5-8.5 end-of-century in this appendix versus −9.5% in Table 5. This is the same left-skew mechanism documented in Appendix F for the underlying productivity response: posterior draws and vessels in which the portfolio collapses pull the median up toward the floor at $\exp(-\beta_f \cdot H_v^{\text{alloc}}) - 1$, while the mean integrates over the long left tail of vessel-and-draw configurations in which the collapse drives f_v^H deep into negative territory and the trip response down with it. Both summaries are valid; we report cross-model medians in the main results table because the median is robust to the extreme tail and is the natural complement to the cross-model inter-quartile range, and we report cross-model means in this appendix because the law of total variance in Equation (13) is constructed on means rather than medians.

This reading is consistent with the underlying biomass-level decomposition reported in Appendix F: for sardine—the stock that drives the artisanal fleet's portfolio collapse—the SSP5-8.5 end-of-century cell exhibits a total SD of 0.010 and an inverted within/between ratio that the appendix explicitly reads as a floor effect at the lower bound of −100%. The fleet-level decomposition reported here inherits that floor effect: the total SD for the artisanal fleet under SSP5-8.5 end-of-century is 0.1804 and the corresponding industrial cell is 0.0374, both of which are small absolute magnitudes and mechanically consistent with the saturated regime.

The substantive policy implication is that, for fleet-level trip projections, the marginal information value of expanding the CMIP6 ensemble is small relative to that of tightening the structural

posterior or refining the description of vessel heterogeneity within fleet. The dominant uncertainty is *not* "which climate model is right" but "given that all of them imply portfolio collapse for the two coastal stocks, how does the resulting heterogeneous economic exposure across vessels translate into trip-level effort." This is the econometric counterpart of the well-known result in the climate-impacts literature that ensemble averaging masks first-order distributional heterogeneity downstream of the climate signal (Cline et al., 2017; Free et al., 2019; Kasperski & Holland, 2013; Oken et al., 2021).

Figure 12 visualises the decomposition.

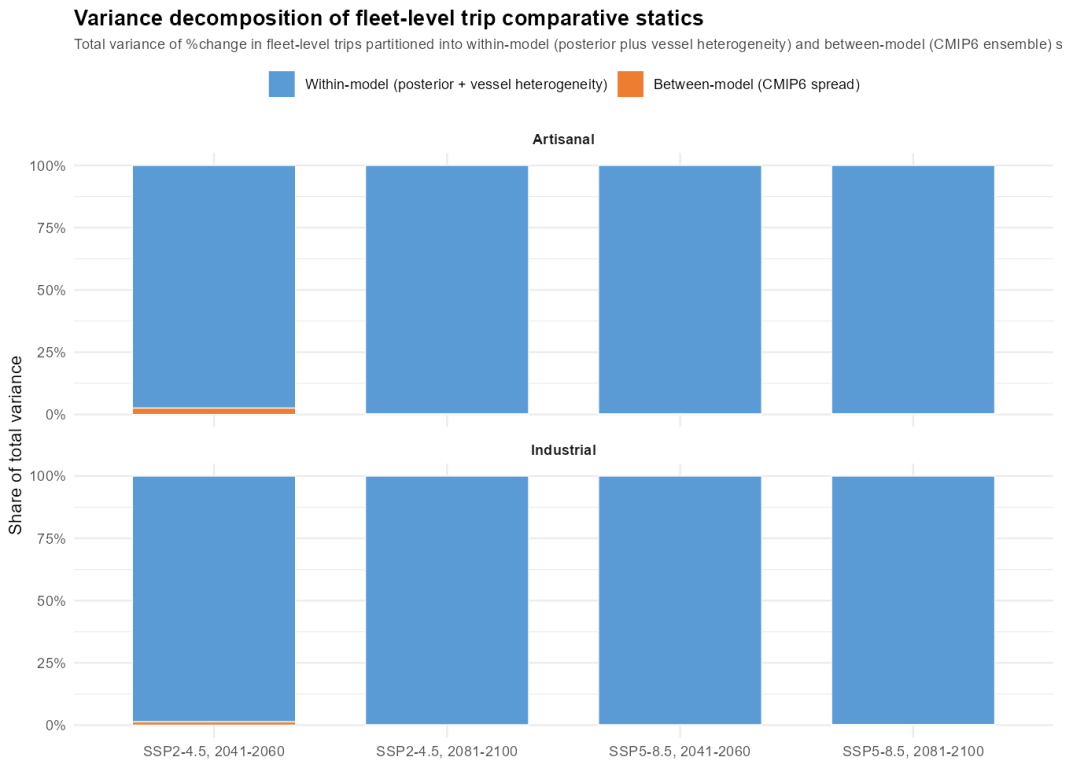

Figure 12: Variance decomposition of $\Delta T_f / T_f^{(0)}$ across the CMIP6 ensemble. Bars stack to one within each scenario–window pair and split the total variance into within-model (posterior pooled with vessel heterogeneity, blue) and between-model (CMIP6 spread, orange) components. The within-model component dominates across all cells, reflecting the floor-effect saturation of the underlying Schaefer steady-state biomass collapse rather than tight cross-model agreement on the climate signal.